

SHETH L.U.J AND SIR M.V. COLLEGE
SUBJECT NAME: DATA VISUALIZATION WITH POWER BI & TABLEAU

Practical No. 6

Aim: Designing Effective and Visually Appealing Dashboards

- a) Apply worksheet-level and workbook-level formatting
- b) Customize number, date, and null value formatting
- c) Design effective color schemes and layouts
- d) Create informative tooltips
- e) Implement “Viz in Tooltip”
- f) Improve readability and user experience
- g) Apply best practices for visual storytelling decide dataset after seeing tis questions

Dataset: Indian E-Commerce Data

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A: Apply Worksheet-Level and Workbook-Level Formatting

Task 1: Connect to the Dataset and Build a Baseline Chart

Opened Tableau Desktop and connected the **List of Orders.csv** file.

The screenshot shows the Tableau Desktop interface with the 'List of Orders.csv' dataset loaded. The sidebar on the left lists various data sources, including 'List of Orders.csv'. The central canvas displays the dataset name and a description. Below the canvas, a data preview table is visible, showing columns for Order ID, Order Date, Customer Name, State, and City. The table contains 560 rows of data.

Order ID	Order Date	Customer Name	State	City
B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
B-25602	01-04-2018	Pearl	Maharashtra	Pune
B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal
B-25604	03-04-2018	Divsha	Rajasthan	Jaipur
B-25605	05-04-2018	Kasheen	West Bengal	Kolkata
B-25606	06-04-2018	Hazel	Karnataka	Bangalore

Dragged 'Order Details.csv' into the canvas and created a relationship using 'Order ID = Order ID'.

The screenshot shows the Tableau Desktop interface with the relationship between 'List of Orders.csv' and 'Order Details.csv' established. The central canvas displays the relationship setup, showing the 'Order ID' field from 'List of Orders.csv' connected to the 'Order ID' field from 'Order Details.csv'. Below the canvas, a data preview table is visible, showing columns for Order ID, Amount, Profit, Quantity, Category, and Sub-Category. The table contains 8 rows of data.

Order ID (Order Details.csv)	Amount	Profit	Quantity	Category	Sub-Category
B-25601	1,275.00	-1,148.00	7	Furniture	Bookcases
B-25601	66.00	-12.00	5	Clothing	Stole
B-25601	8.00	-2.00	3	Clothing	Hankerchief
B-25601	80.00	-56.00	4	Electronics	Electronic Games
B-25602	168.00	-111.00	2	Electronics	Phones
B-25602	424.00	-272.00	5	Electronics	Phones
B-25602	2,617.00	1,151.00	4	Electronics	Phones
B-25602	561.00	212.00	3	Clothing	Saree
B-25602	119.00	-5.00	8	Clothing	Saree

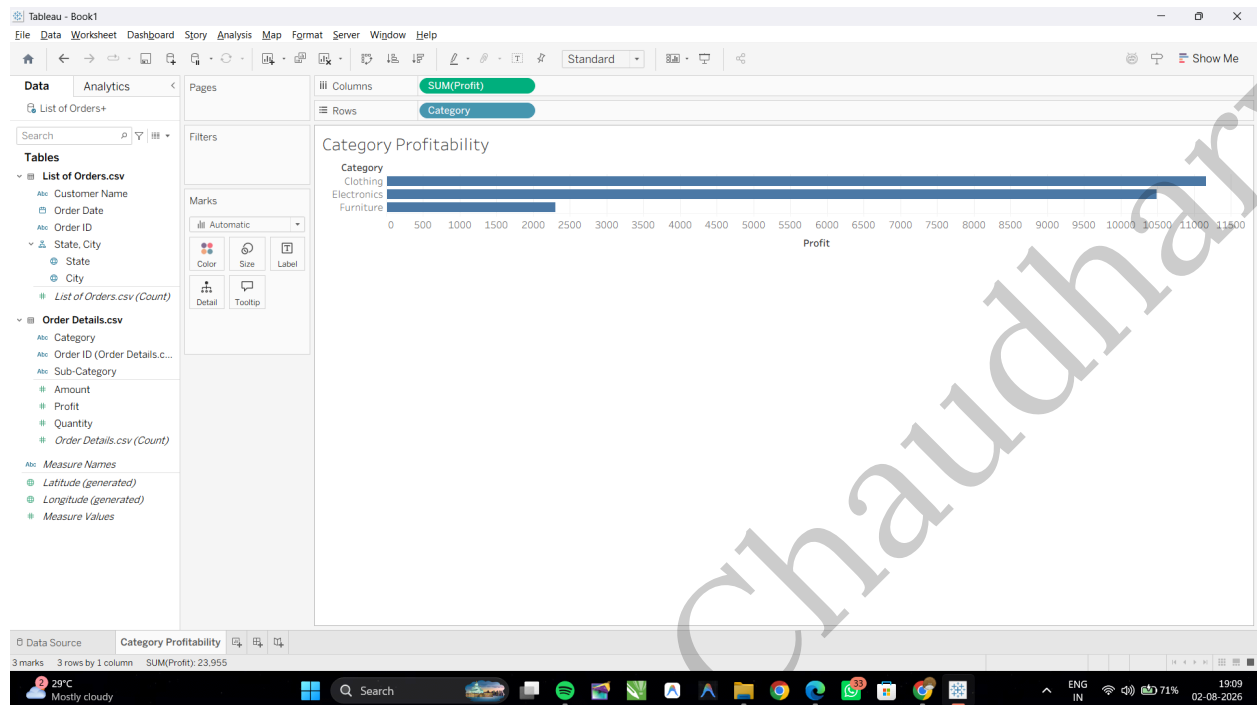
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Opened Sheet 1, renamed it to *Category Profitability*, and created a horizontal bar chart using *Category* on Rows and *Profit* on Columns.

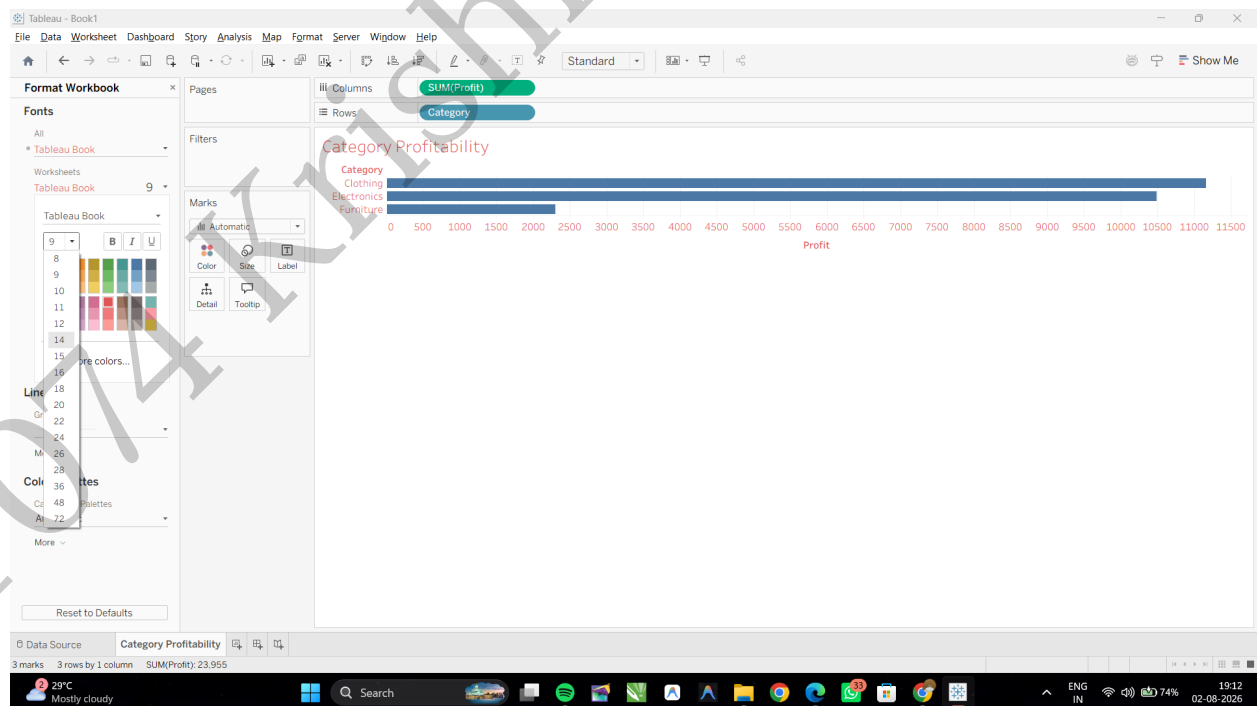
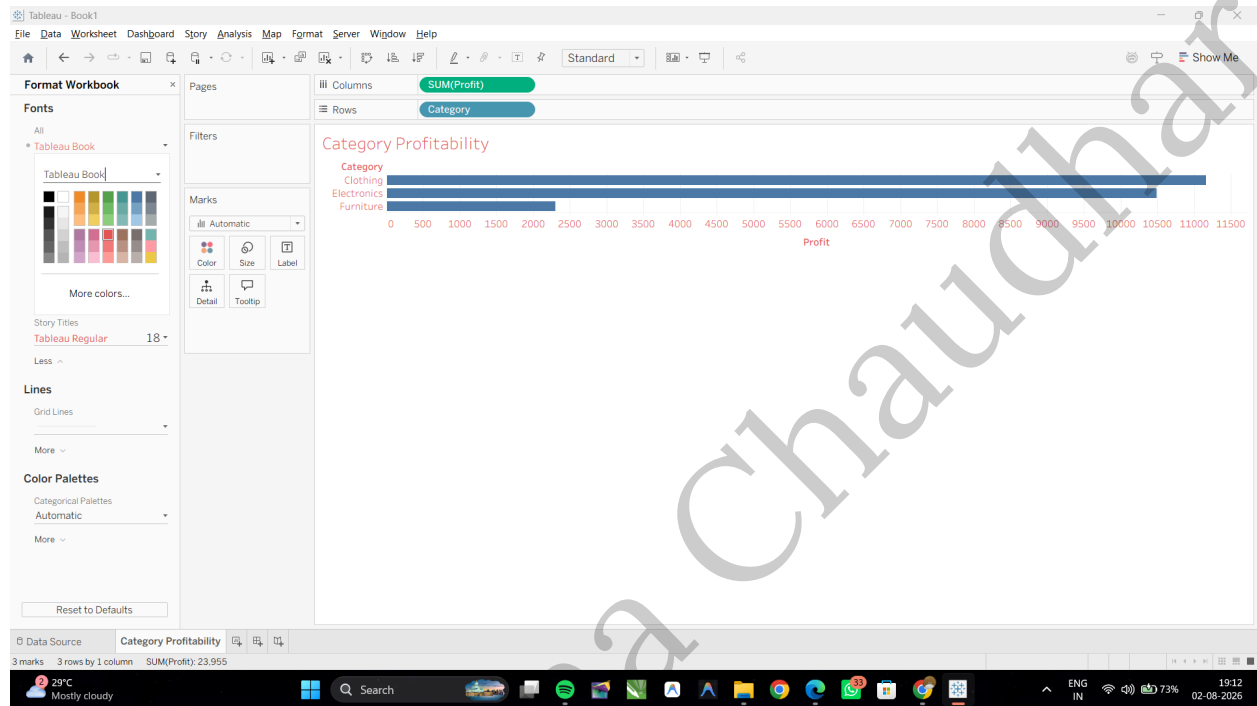


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Task 2: Apply Workbook-Level Formatting

Formatted the Workbook: Set the workbook font to Segoe UI/Tableau Book, customized the worksheet title to Segoe UI Semibold (14 pt, Dark Gray), turned off grid lines, and closed the Format Workbook pane.

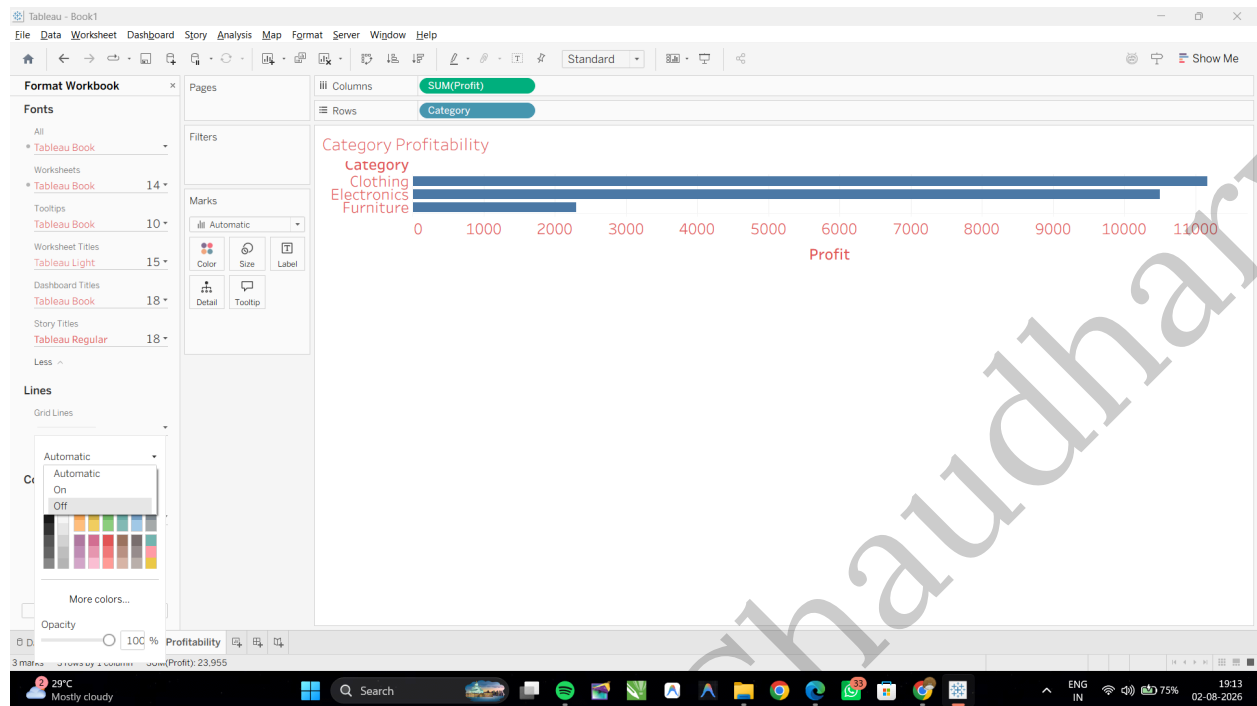


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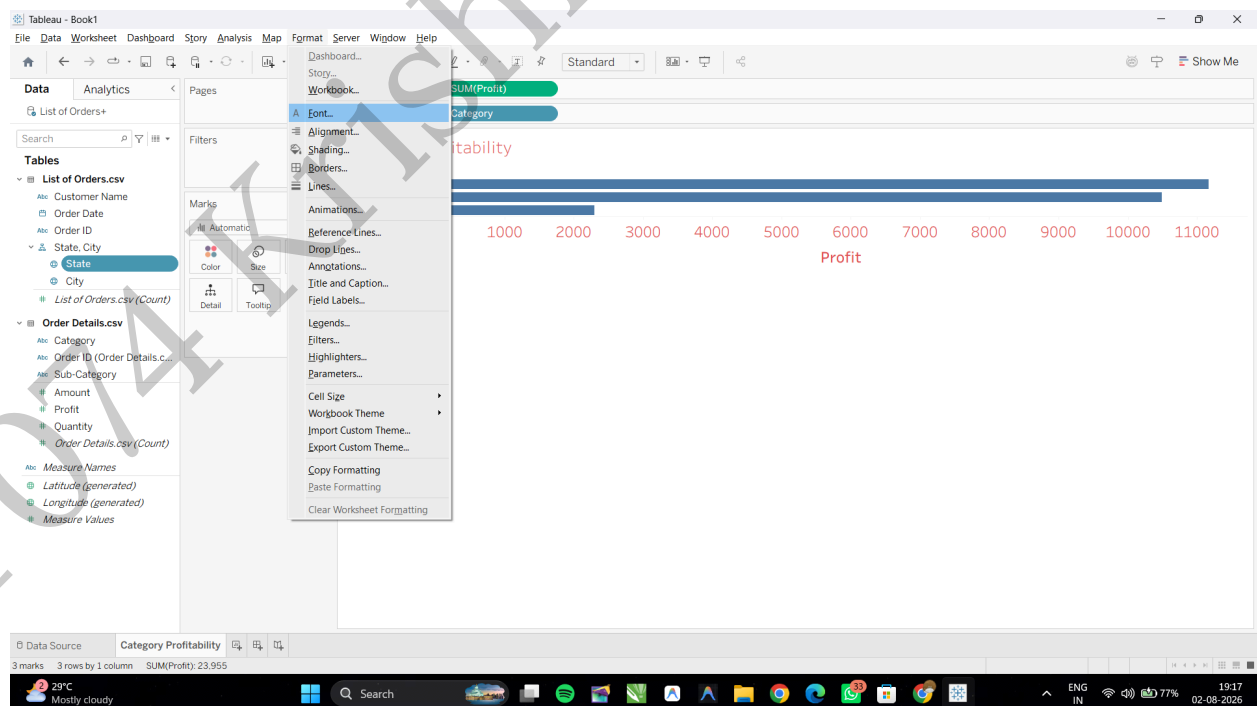
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Task 3: Apply Worksheet-Level Formatting

Formatted the Worksheet: Set the header and pane fonts, removed row and column dividers, and formatted the zero line as solid dark gray.

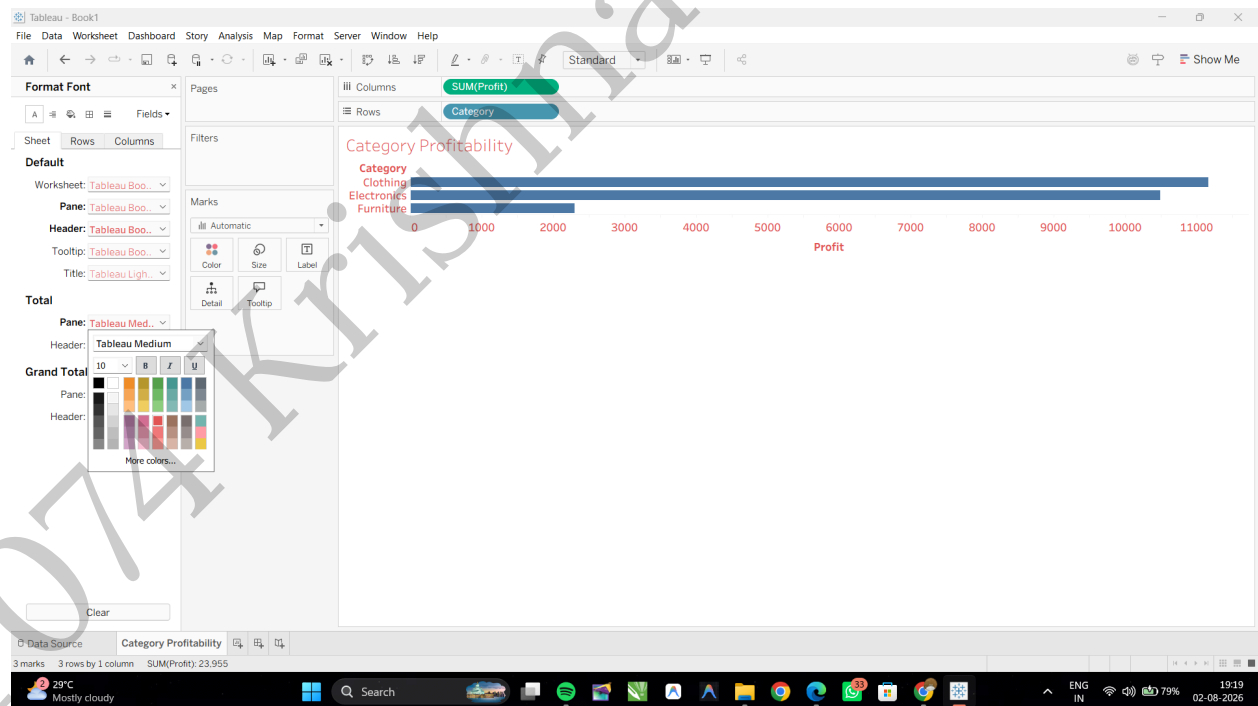
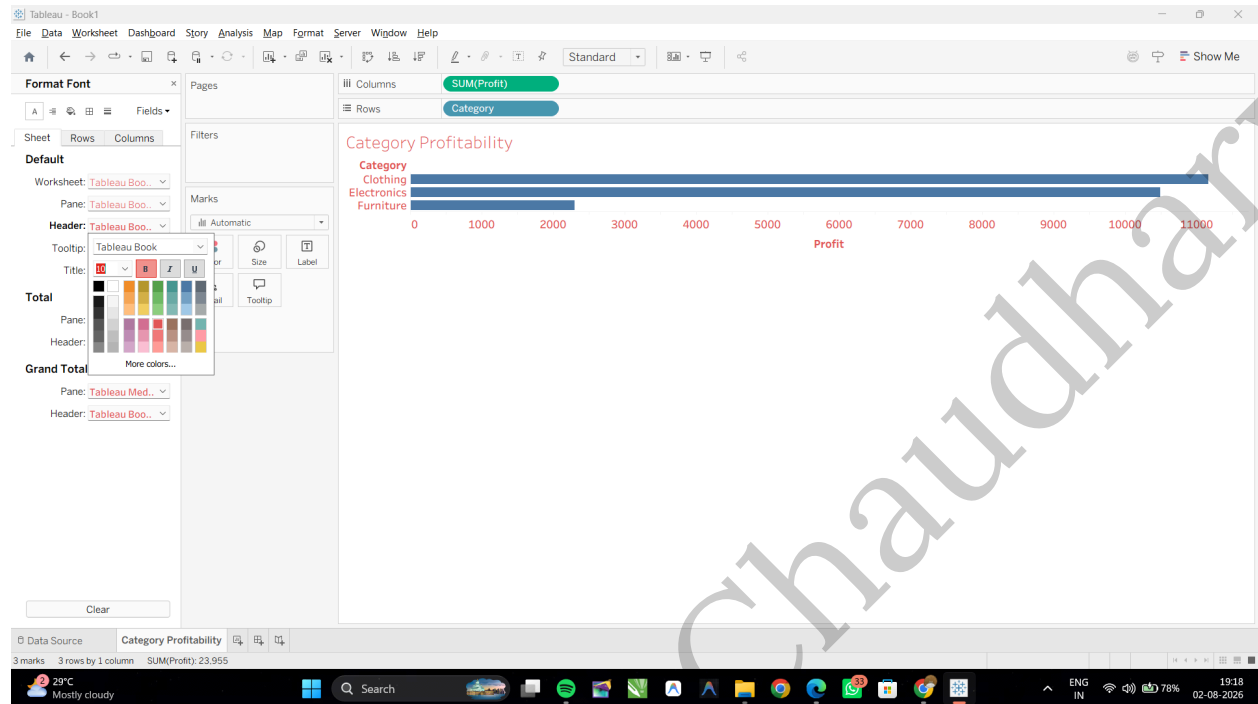


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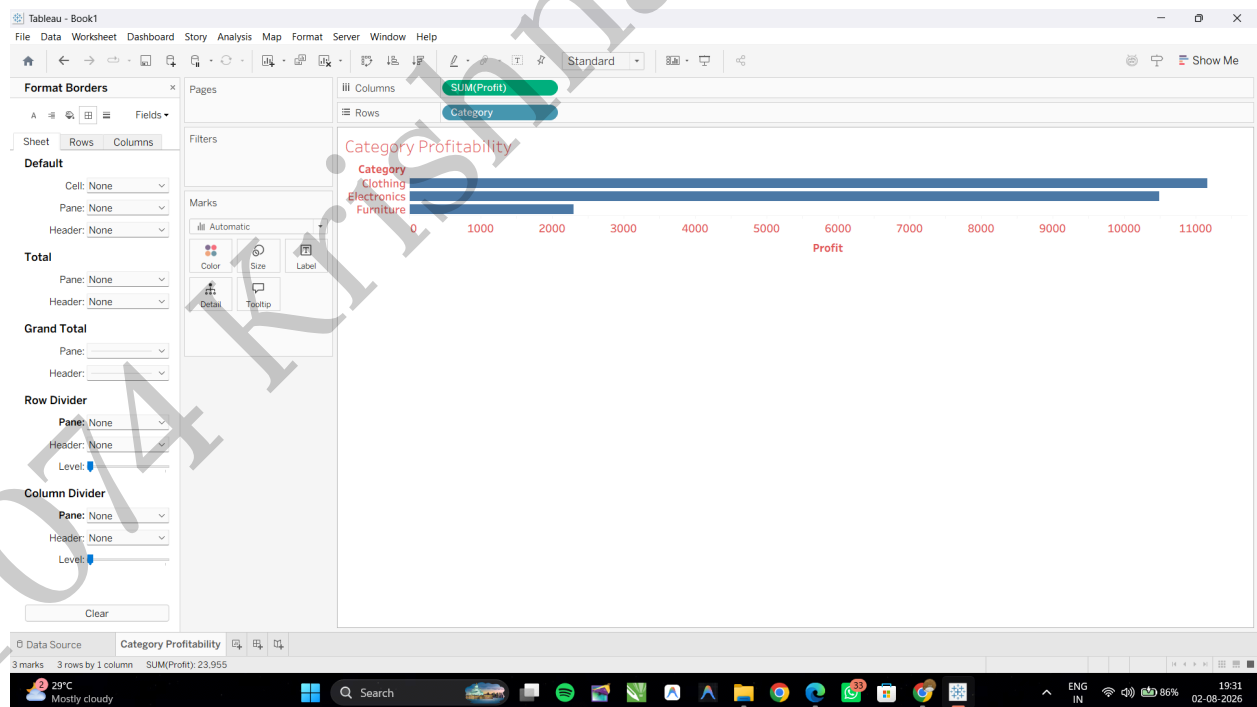
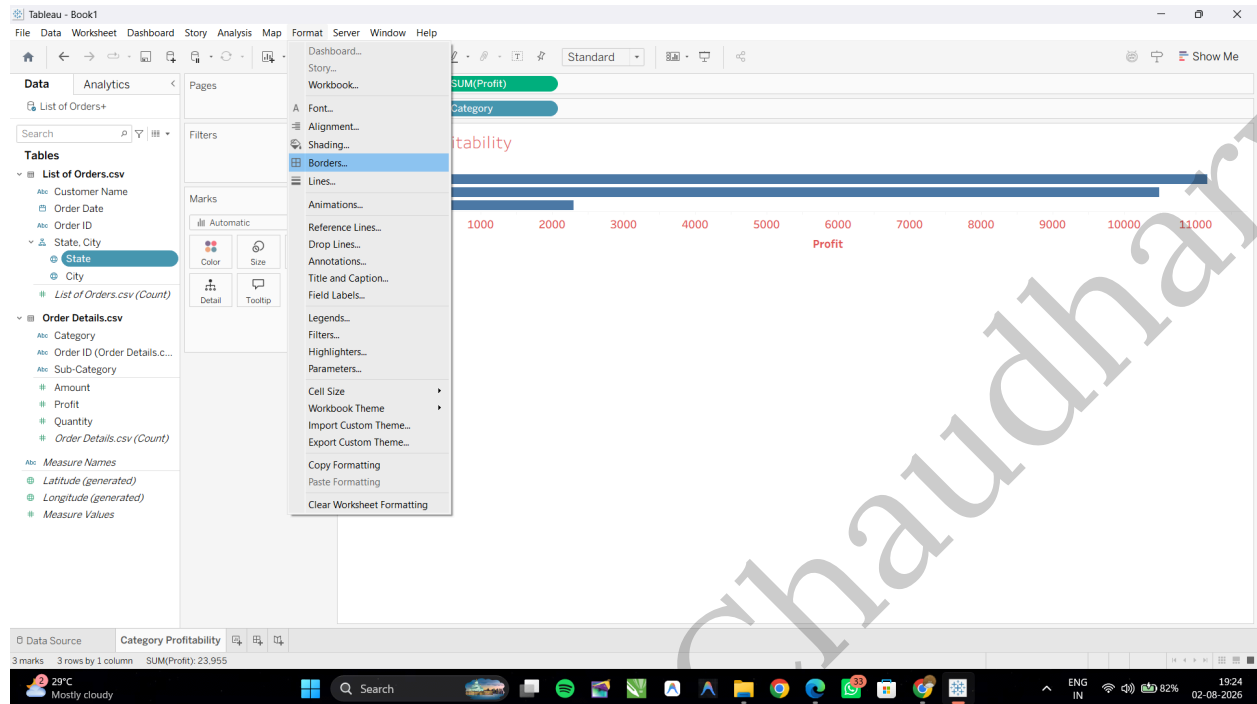
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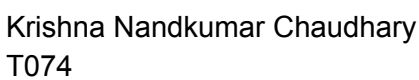
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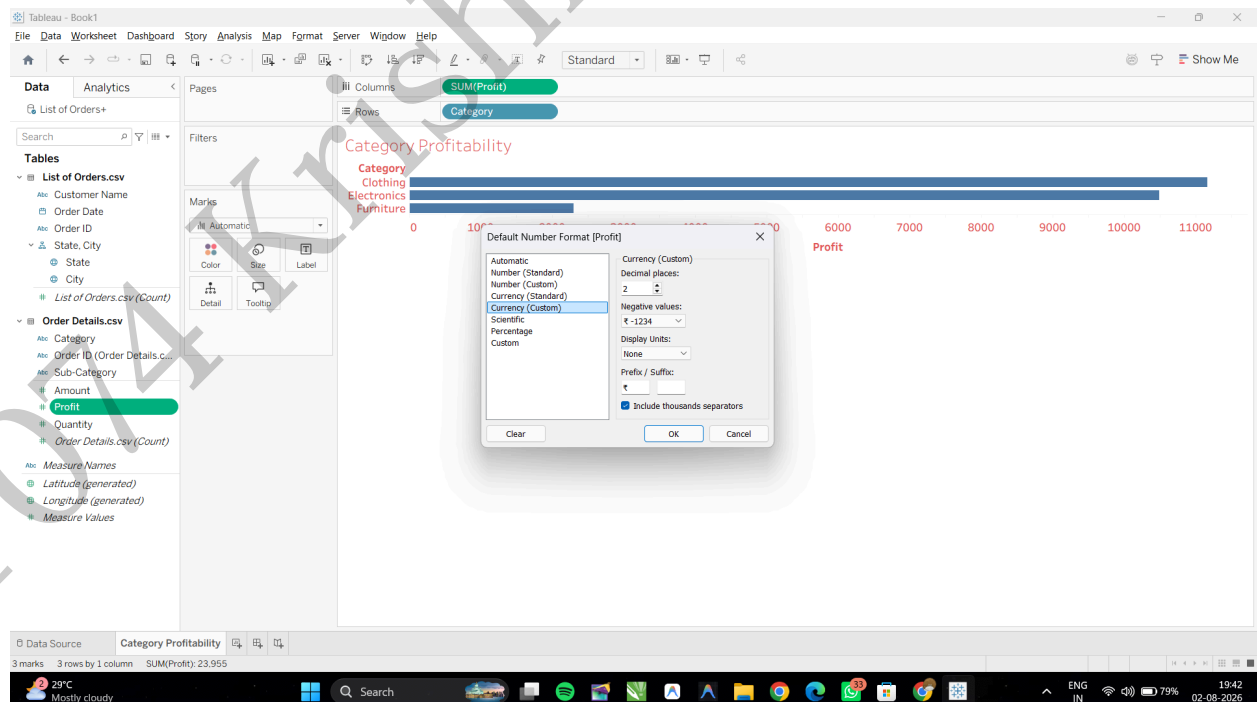
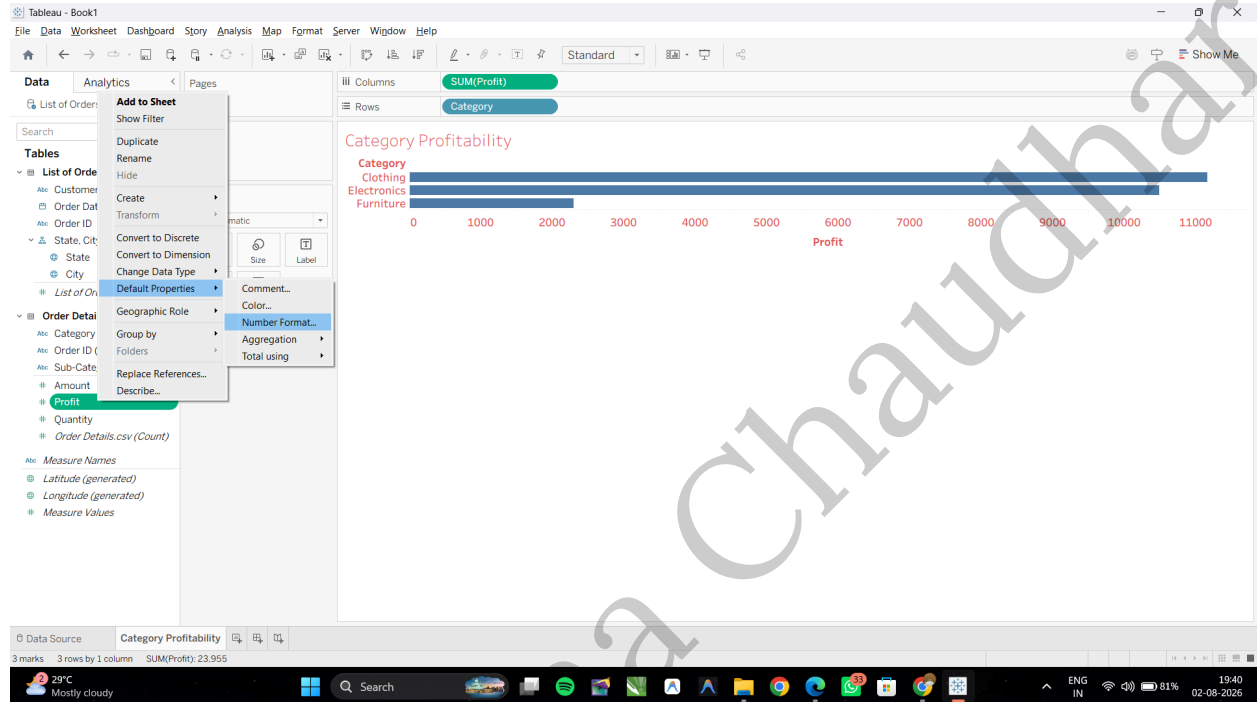
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B: Customize Number, Date, and Null Value Formatting

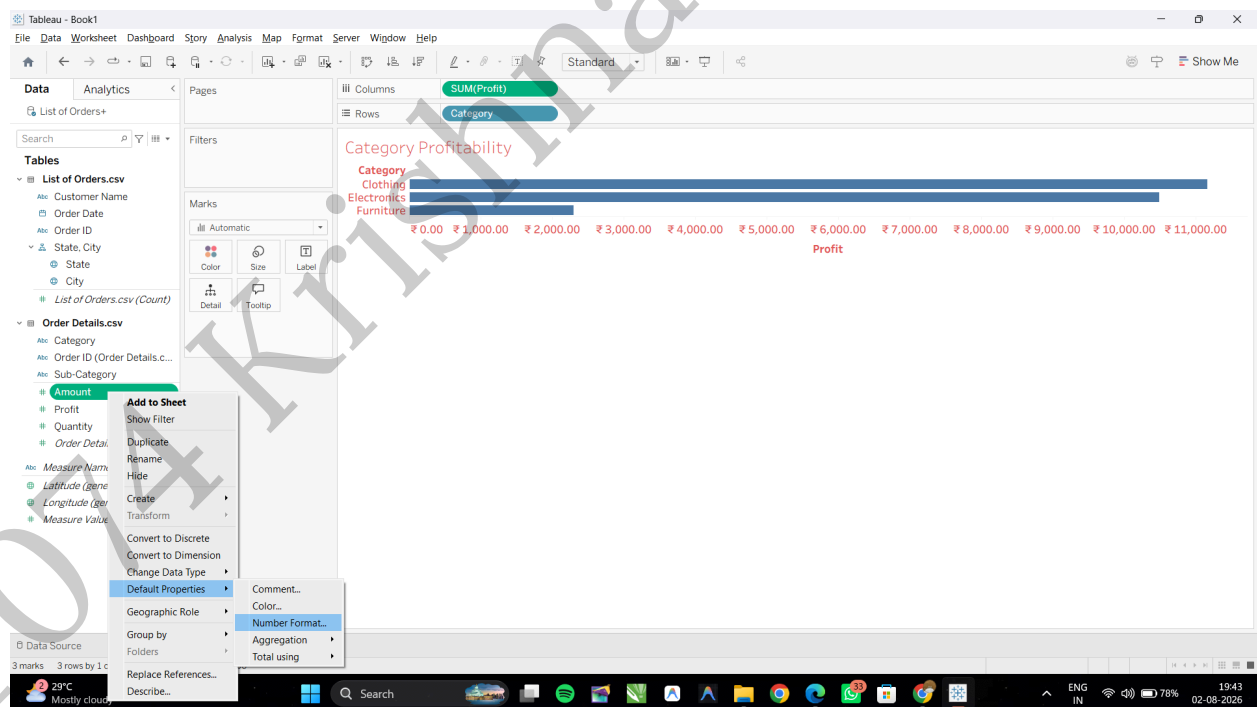
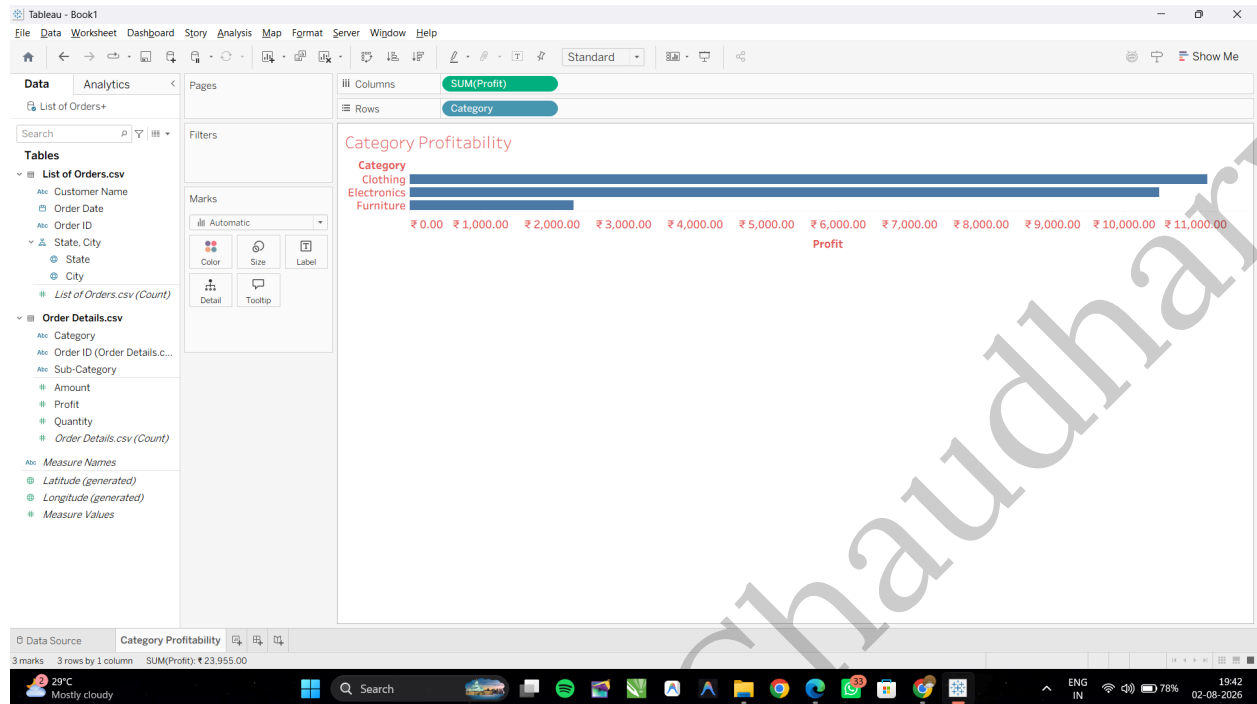
Task 1: Format Profit and Amount as Indian Currency

Right-clicked Profit, selected Default Properties → Number Format, set Currency (Custom) with ₹ as the prefix and 0 decimal places, then repeated the same formatting for Amount.



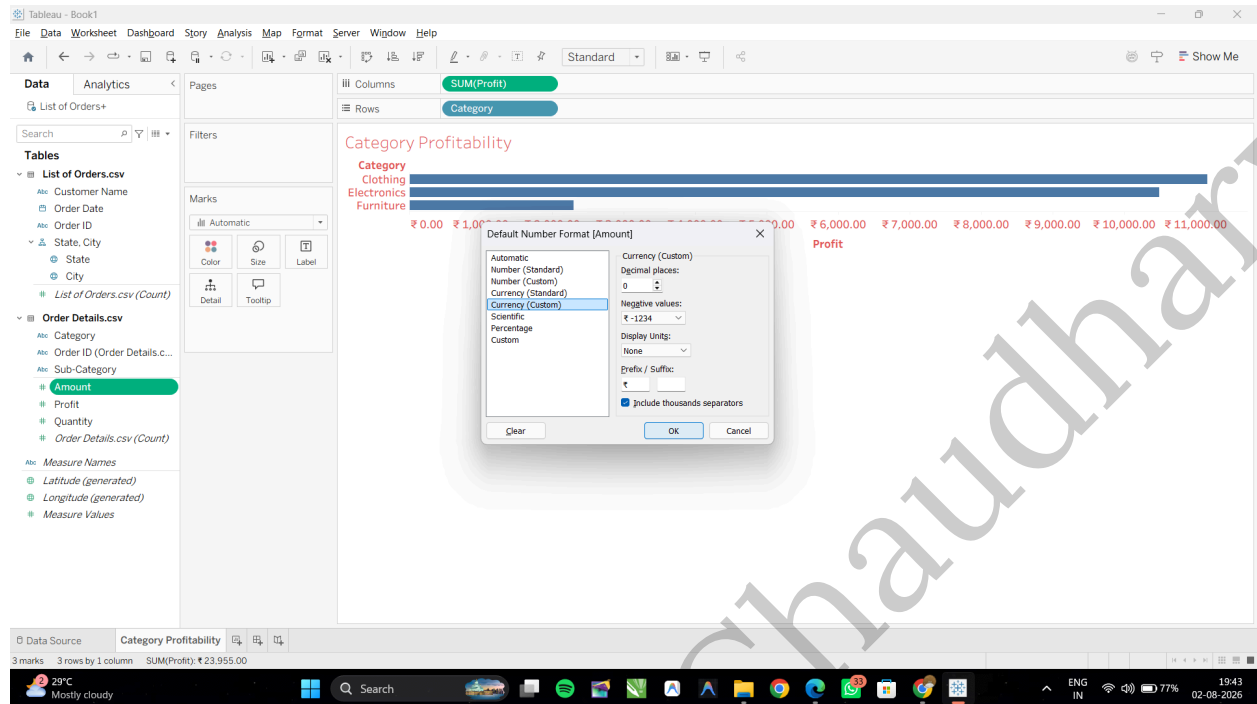
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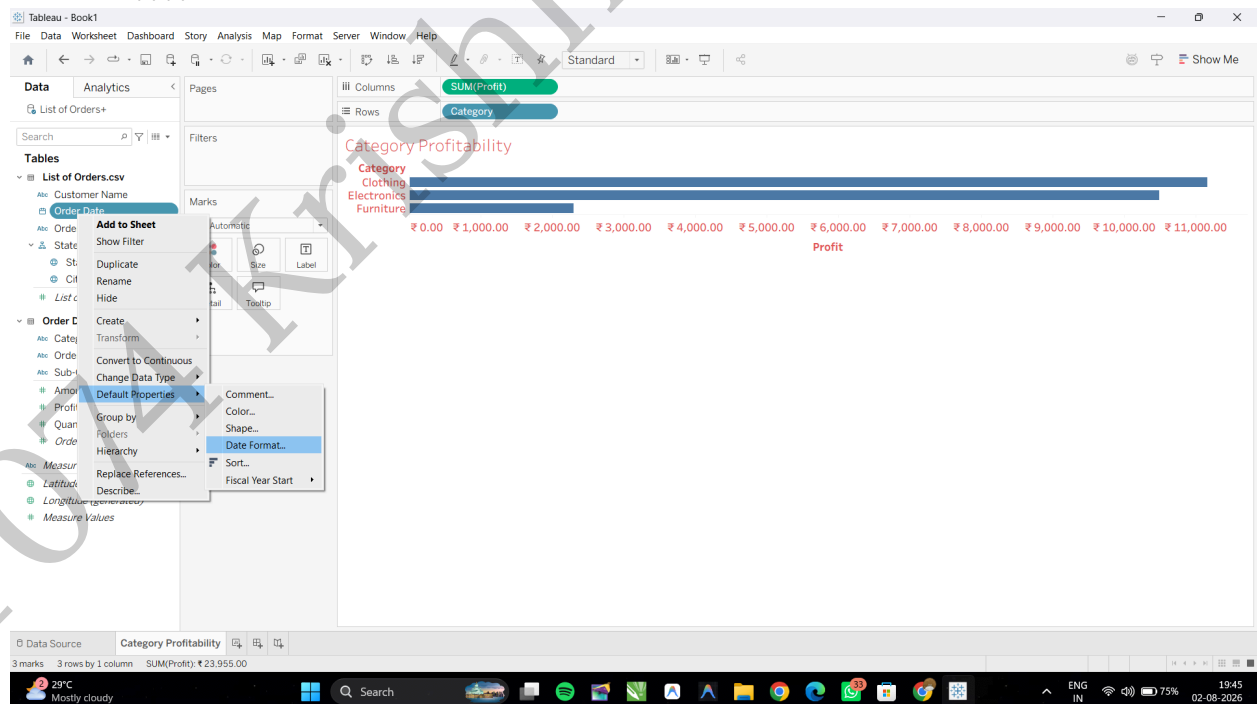
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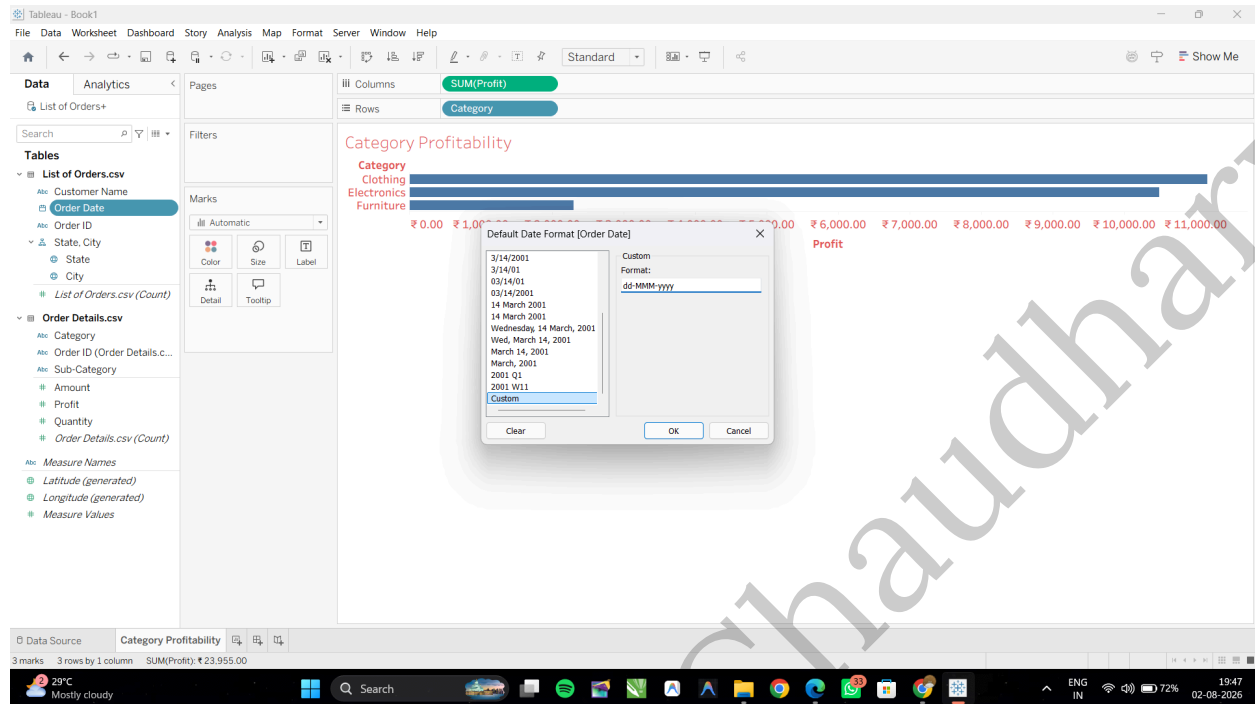
Task 2: Standardize Date Formatting

Right-clicked Order Date, selected Default Properties → Date Format, and set the format to dd-MMM-yyyy.



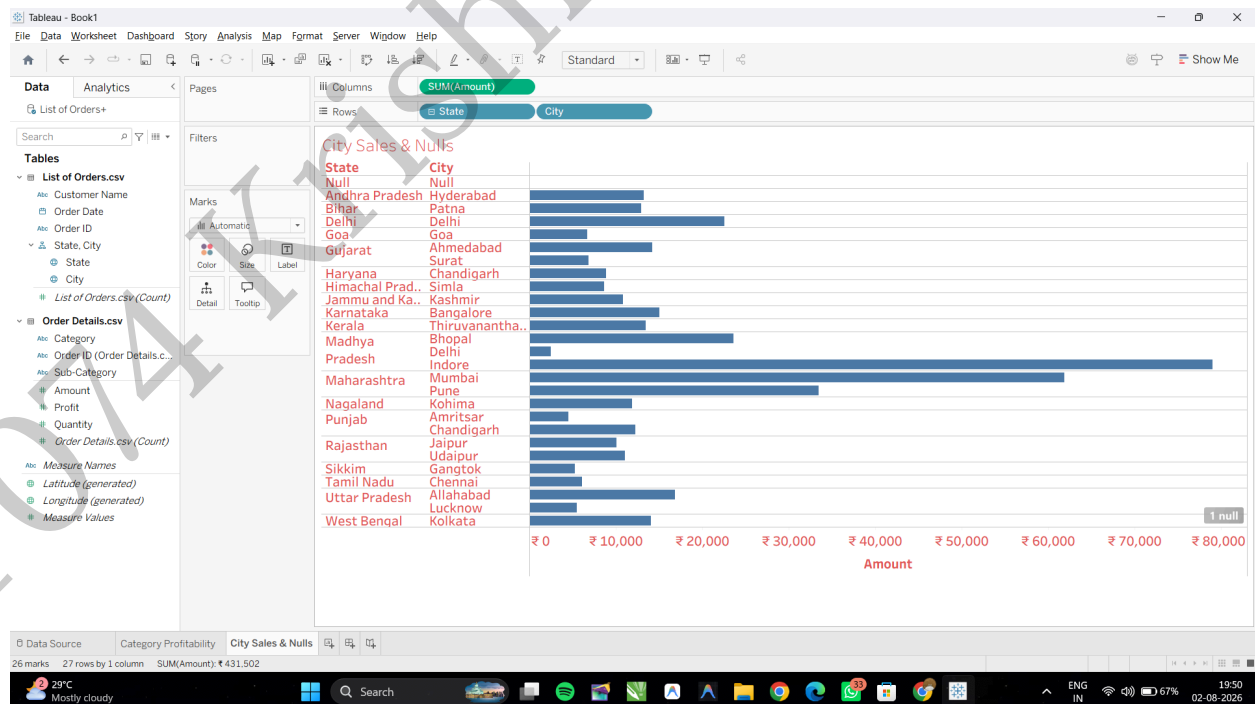
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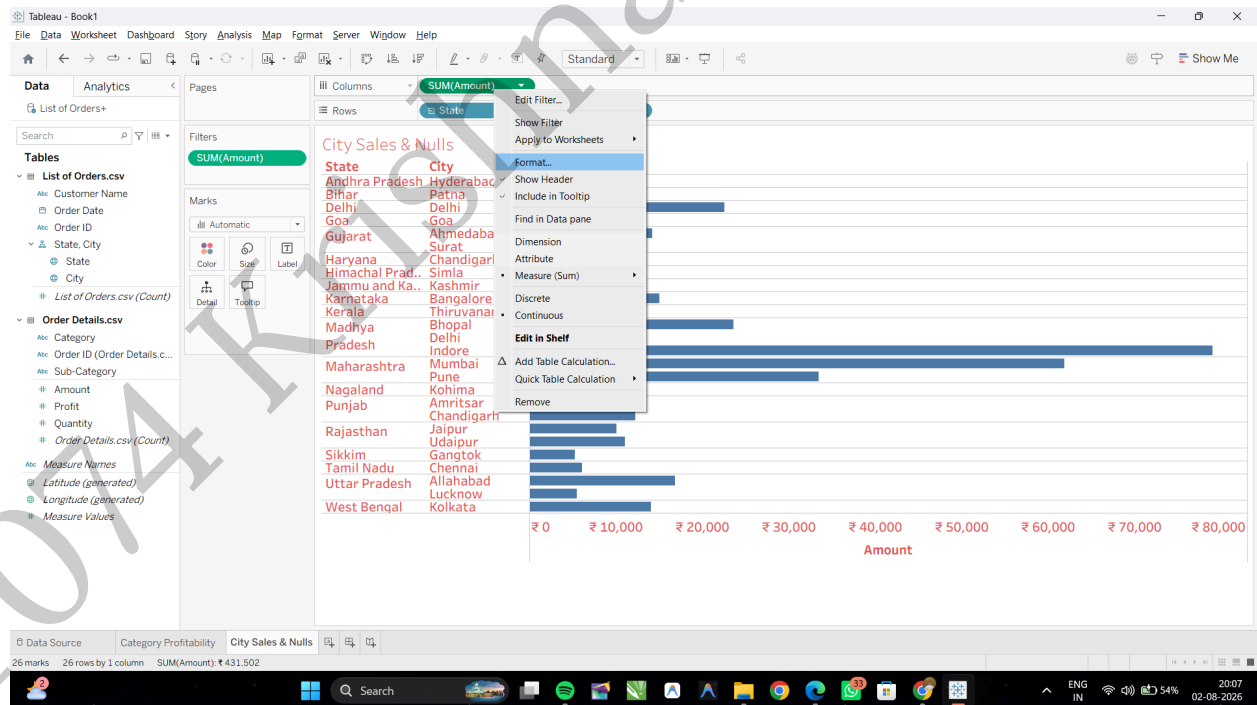
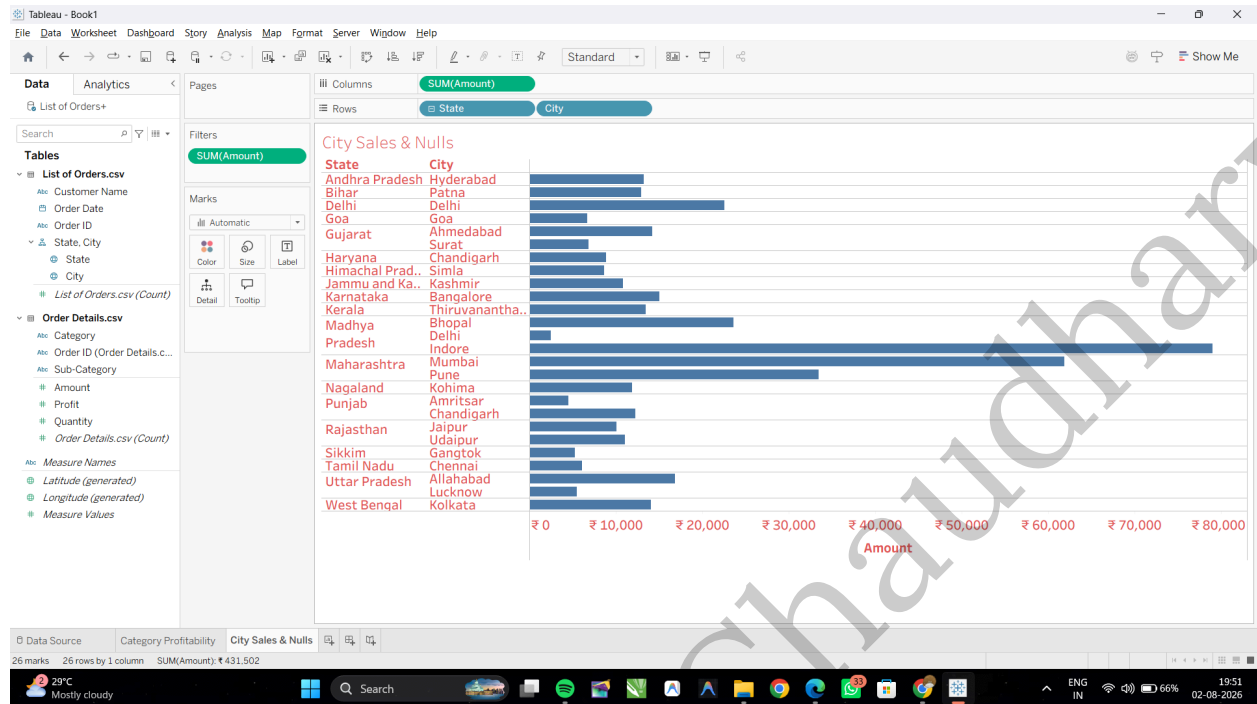
Task 3: Handle and Custom-Format Null Values

Created a new worksheet named **City Sales & Nulls**, dragged **State** and **City** to the **Rows** shelf, **Amount** to the **Columns** shelf, checked for NULL values, and customized the **Special Values** text to **Not Reported / N/A**.

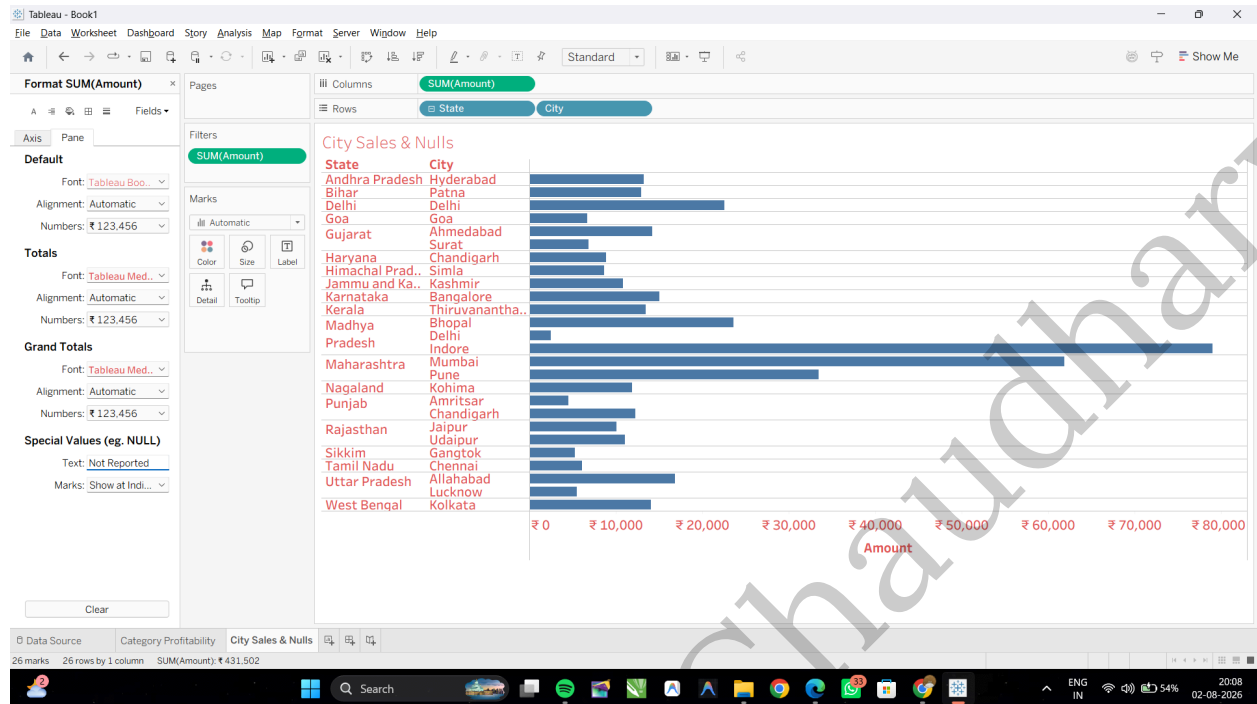


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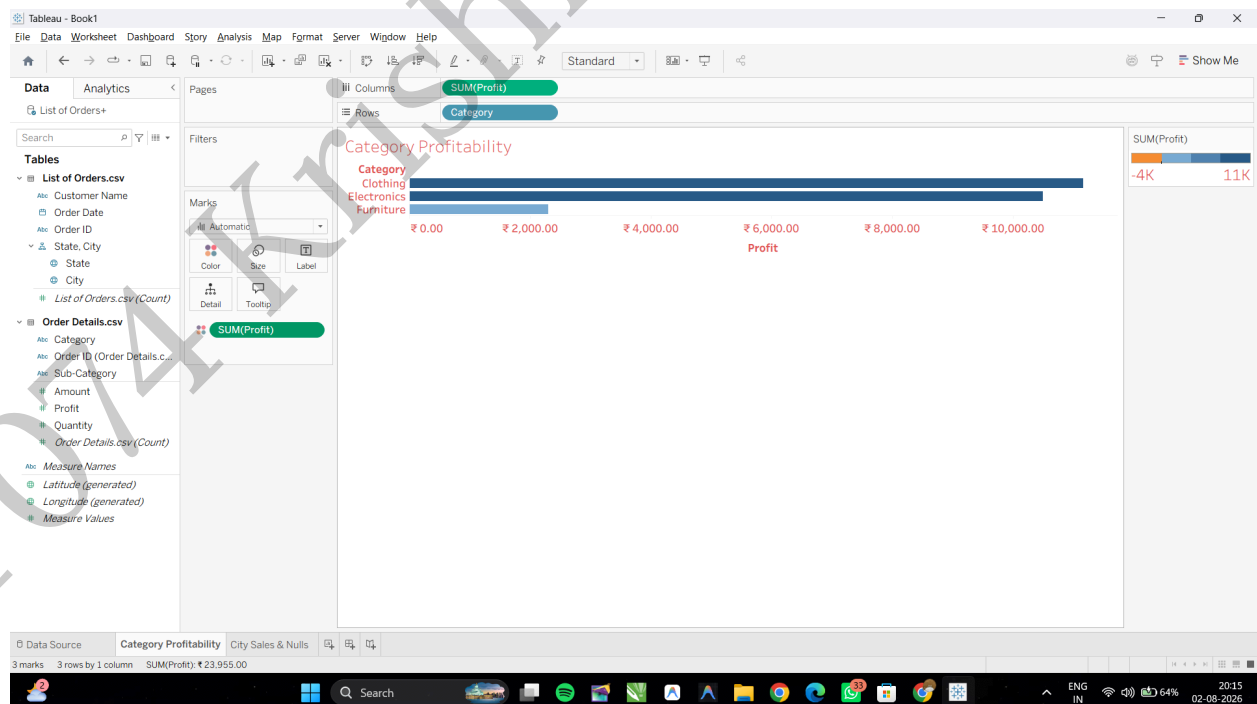
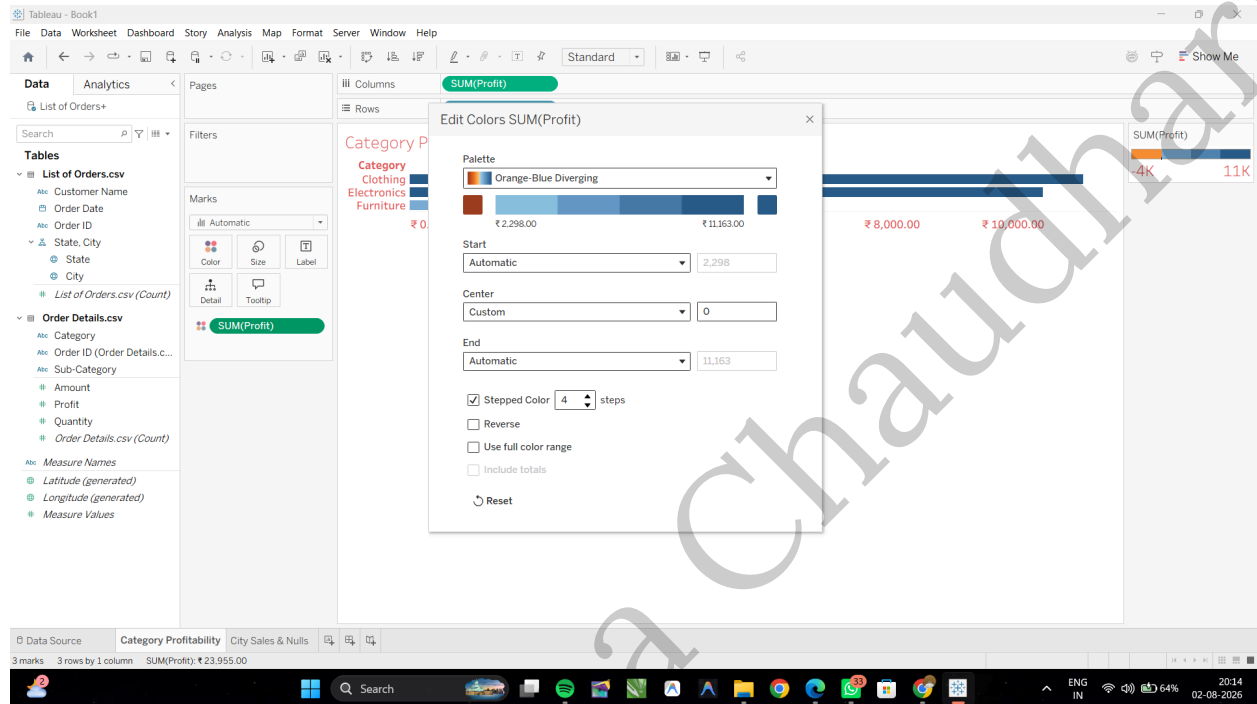
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C: Design Effective Color Schemes and Layouts

Task 1: Apply a Diverging Colour Scheme

Opened the **Category Profitability** worksheet, dragged **Profit** to the **Color** card, opened **Edit Colors**, selected the **Orange-Blue Diverging** palette, enabled **Stepped Colors**, set the **Center** value to **0**, and applied the changes.



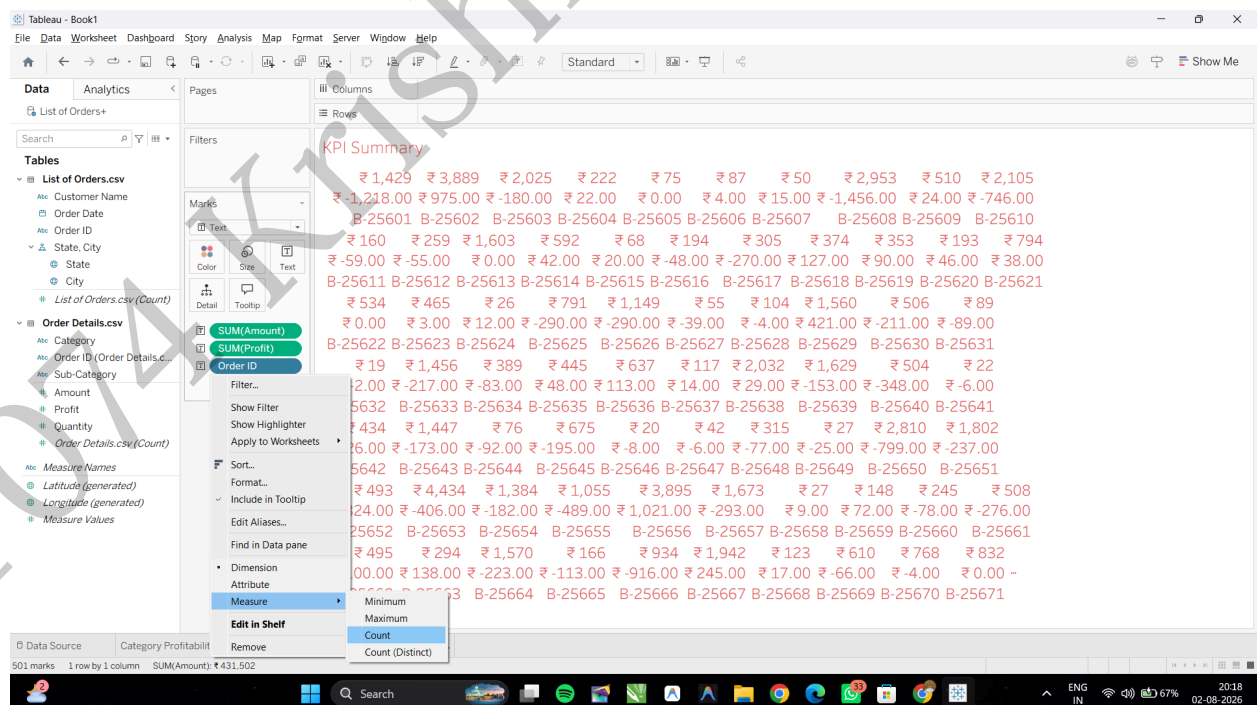
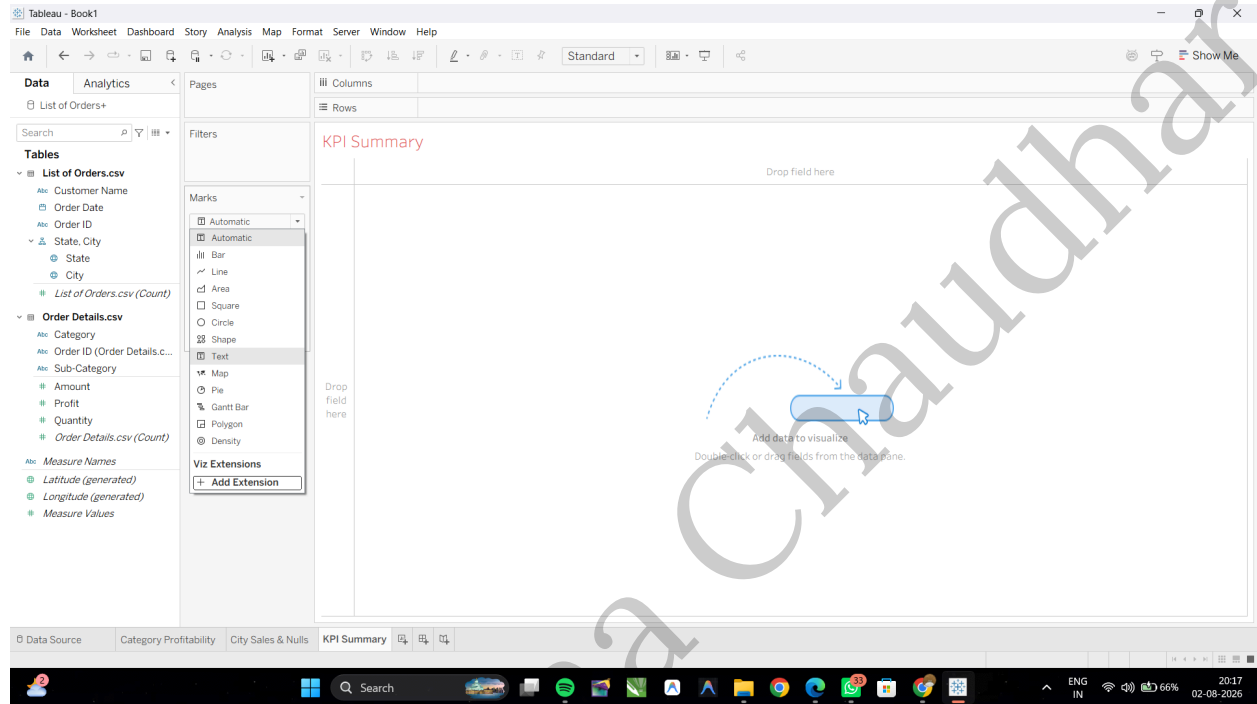
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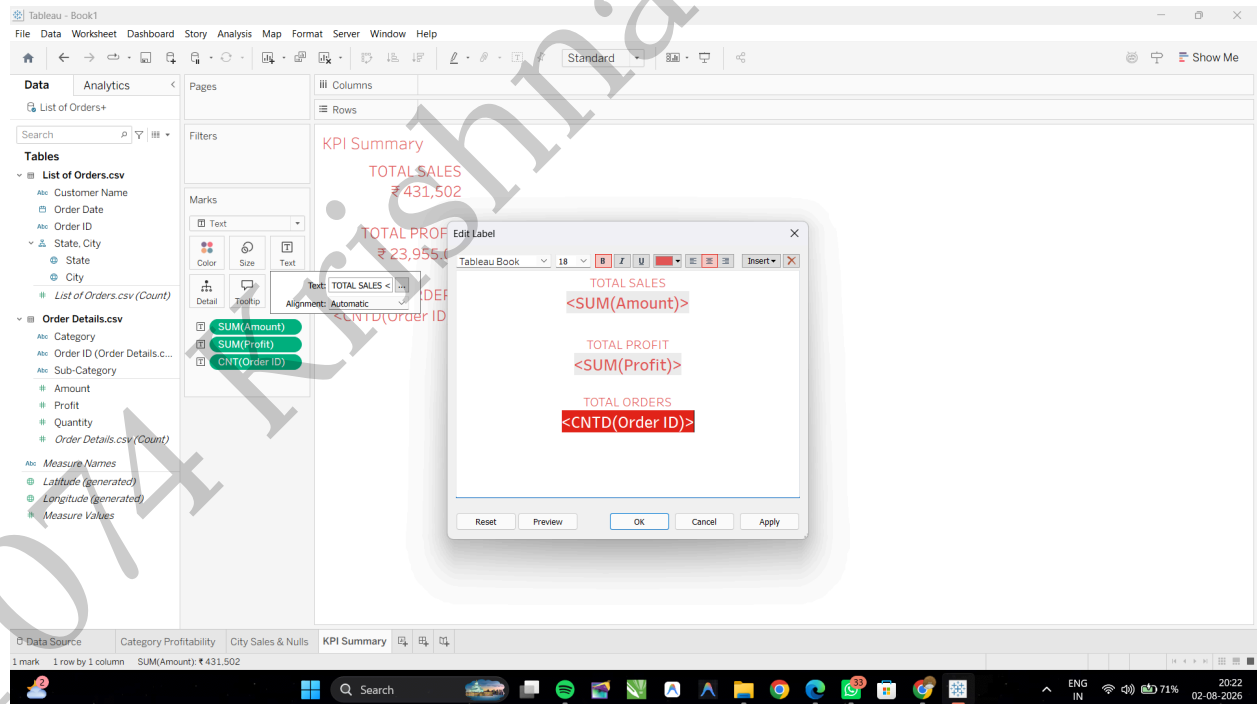
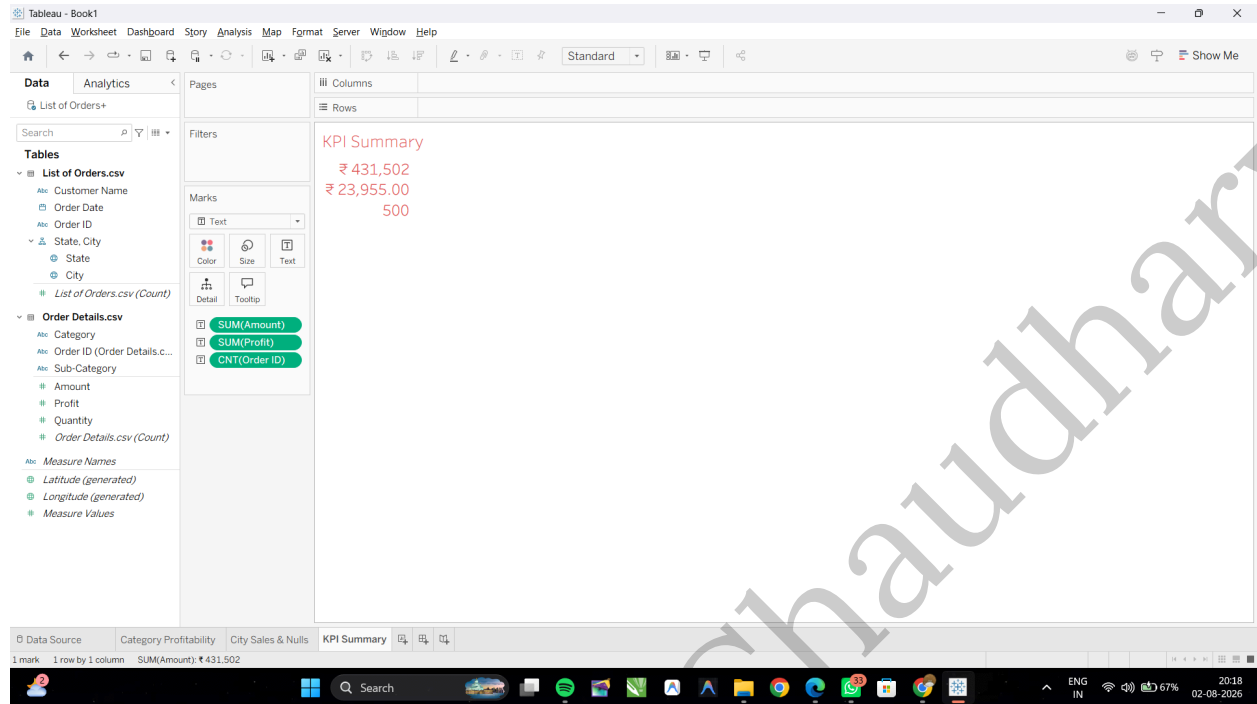
Task 2: Build KPI Summary Sheet

Created a worksheet named **KPI Summary**, changed the **Mark Type** to **Text**, dragged **Amount**, **Profit**, and **Order ID** (set to **Count (Distinct)**) to **Text**, created KPI labels (**Total Sales**, **Total Profit**, and **Total Orders**), formatted the labels and values, and centered the KPI text.



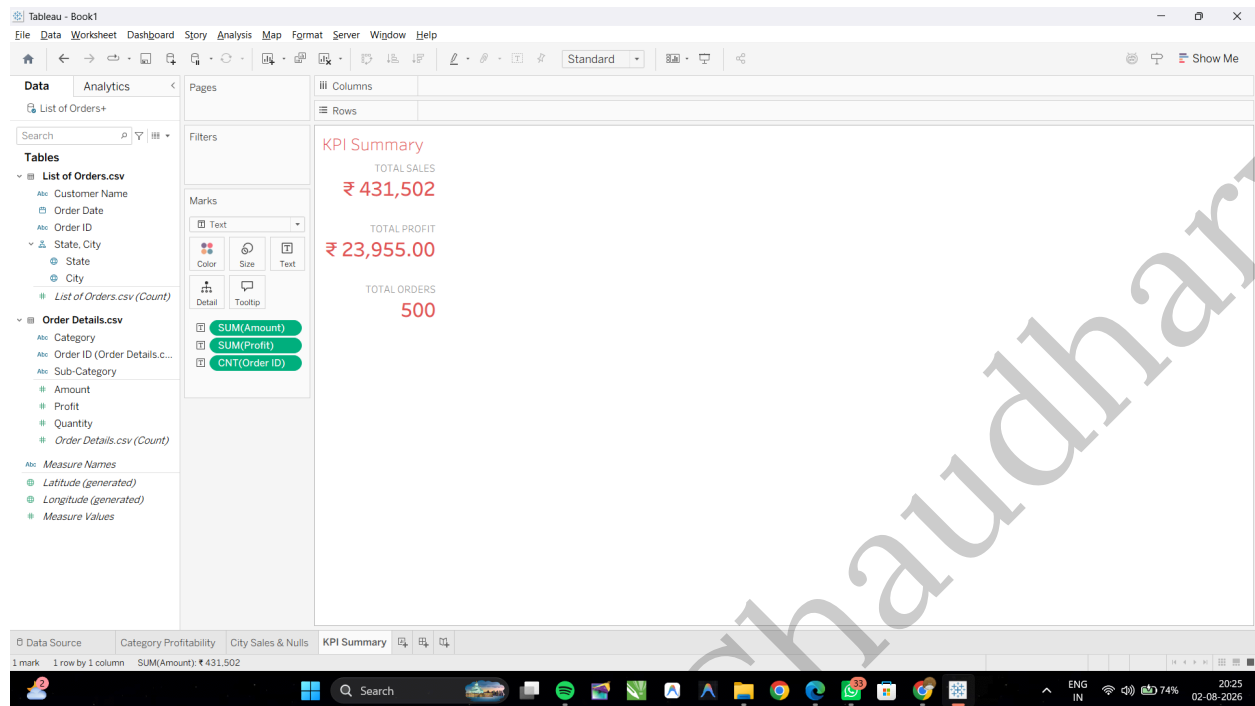
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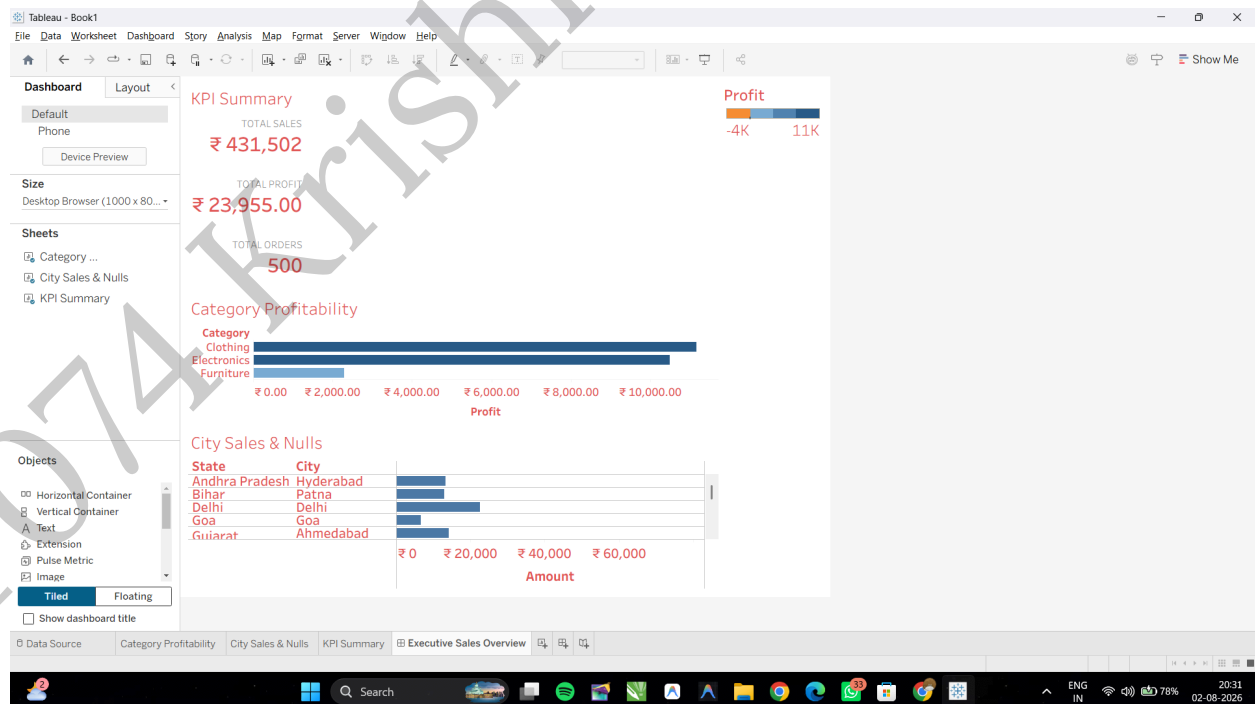
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Task 3: Create Dashboard Container Layout

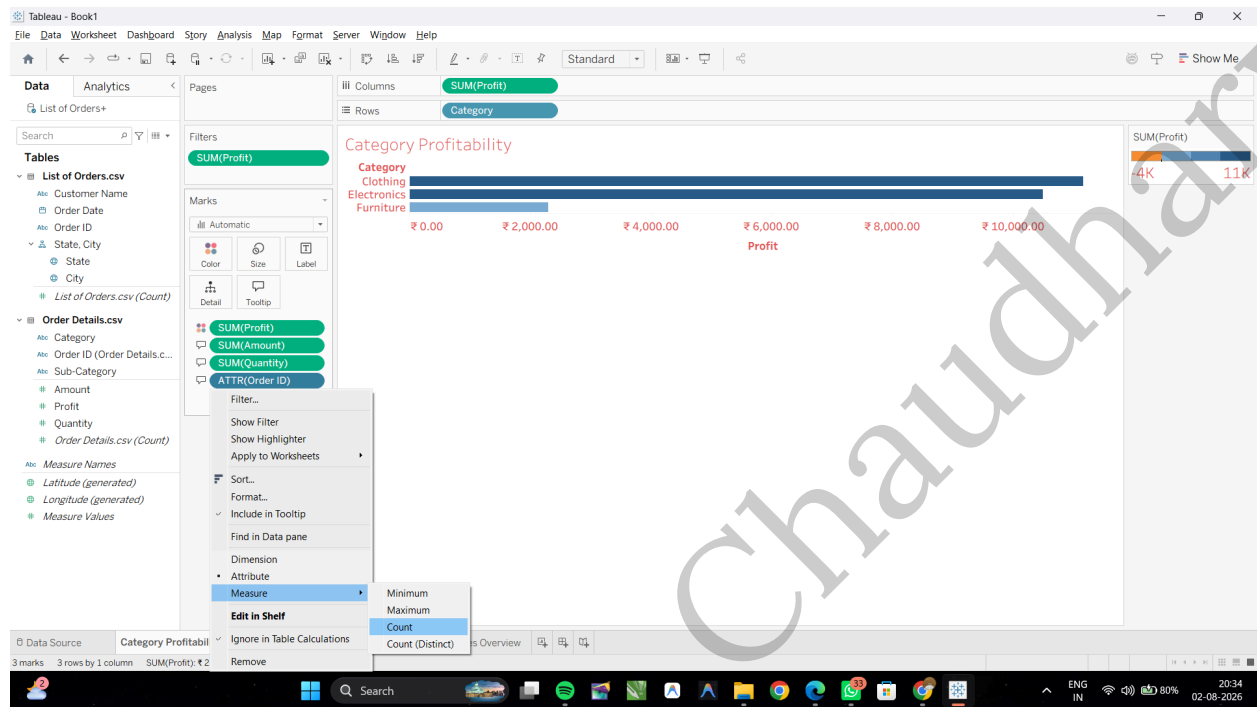
Created a new dashboard, renamed it **Executive Sales Overview**, set the dashboard size to **1000 × 800**, added a **Vertical Container**, placed the **KPI Summary**, **Category Profitability**, and **City Sales & Nulls** worksheets in order, and removed unnecessary titles where required.



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D: Create Informative Tooltips

Task 1: Open Tooltip Editor

Opened the **Tooltip** option from the **Marks** card.

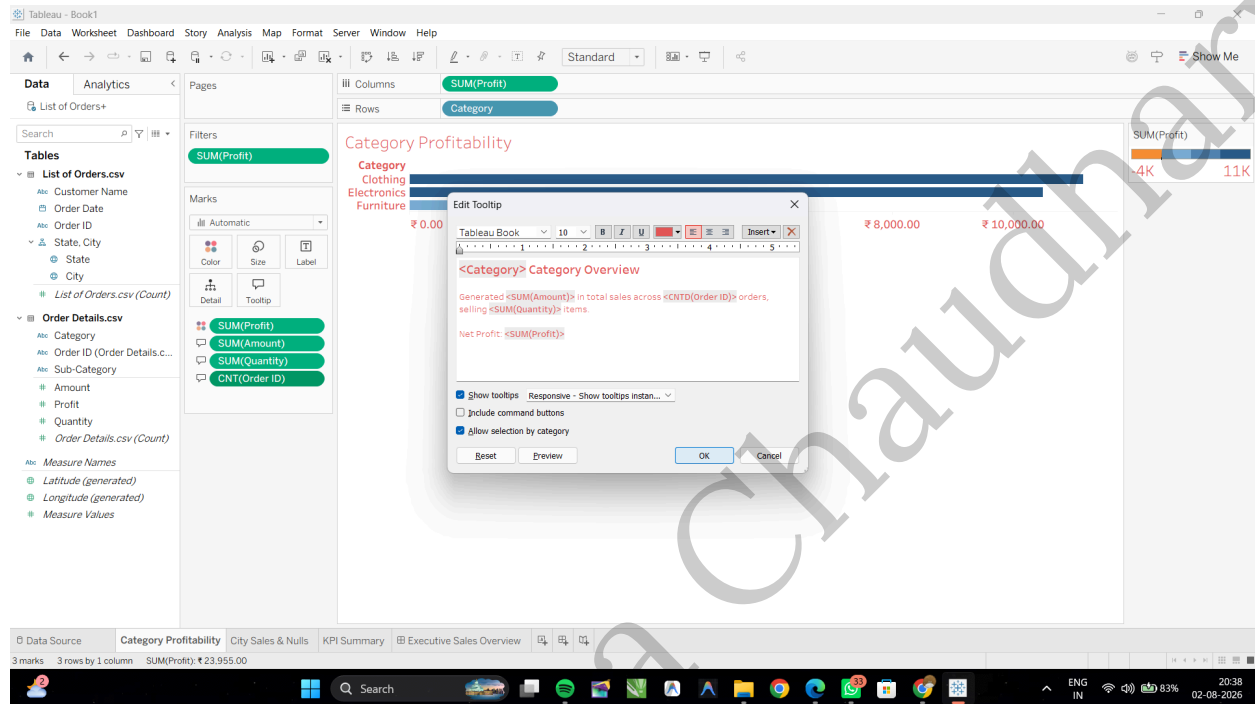


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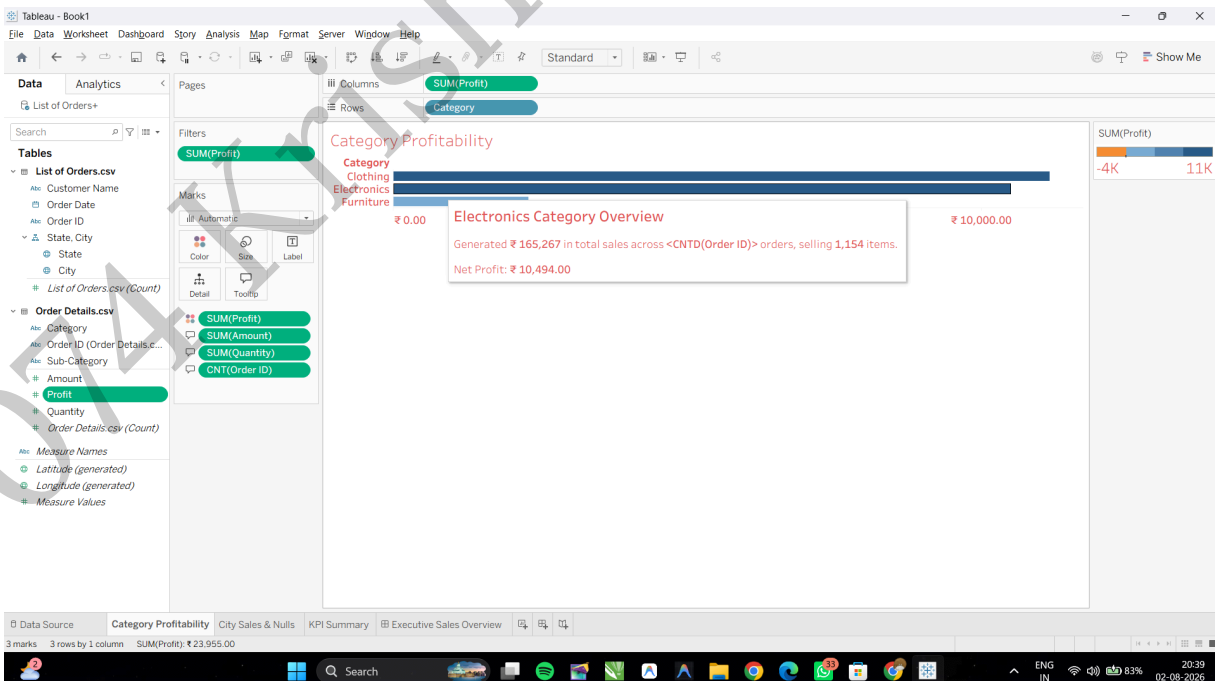
Task 2, 3: Add Contextual Measures and Create Sentence-Based Tooltip

Added **Amount**, **Quantity**, and **CNTD(Order ID)** to the **Tooltip**, deleted the default tooltip, created a custom sentence-based tooltip displaying **Category Overview**, **Total Sales**, **Total Orders**, **Quantity Sold**, and **Net Profit**, formatted the heading, body, and dynamic values, and disabled **Include Command Buttons**.



Task 4: Test Tooltip

Hovered over the bars and verified the custom tooltip.



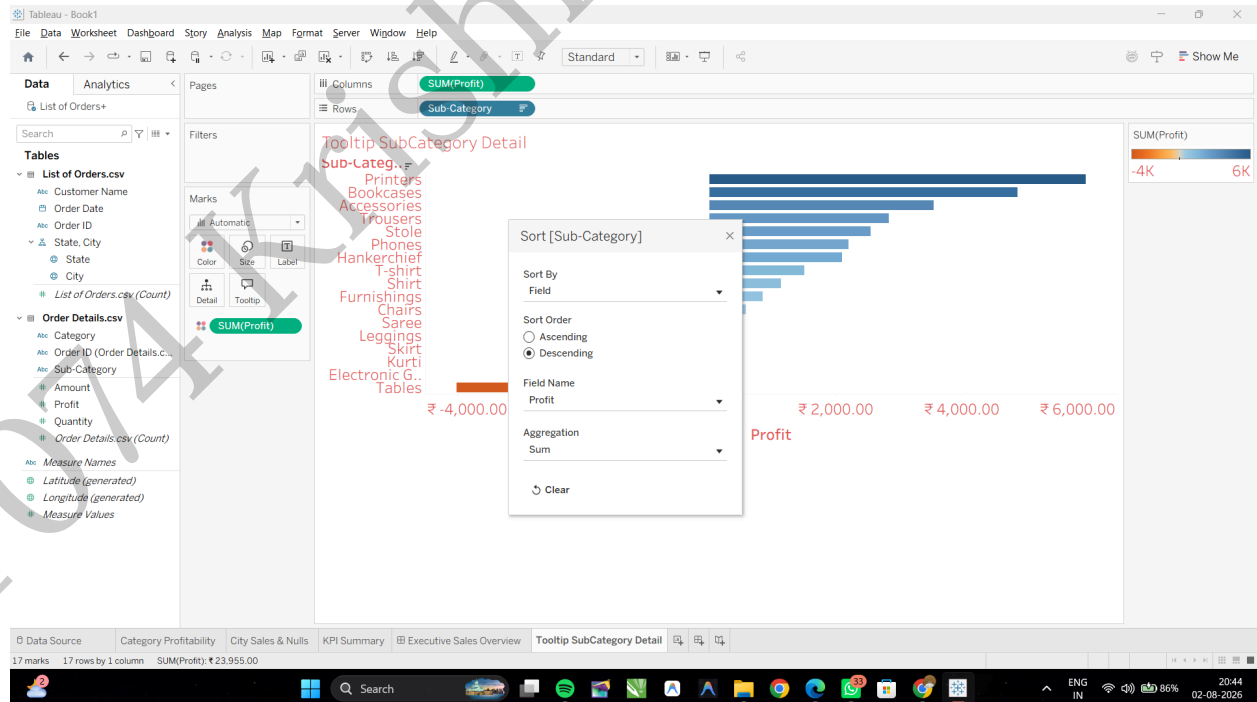
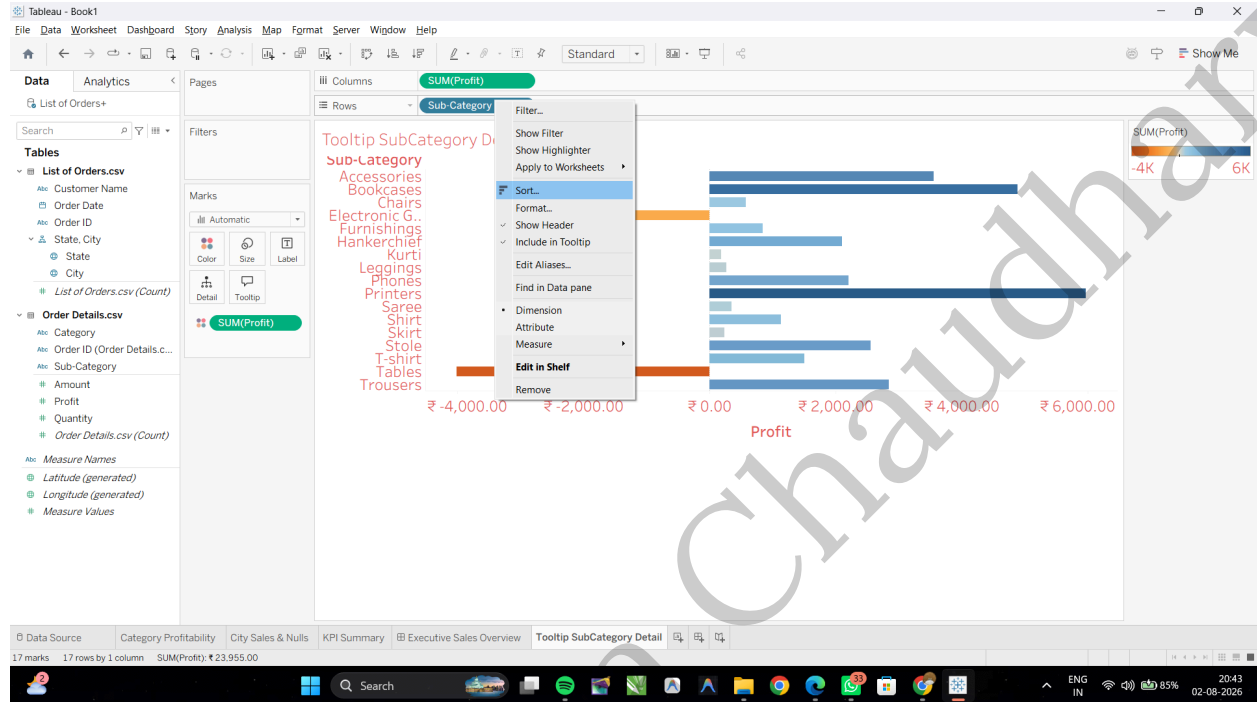
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E: Implement "Viz in Tooltip"

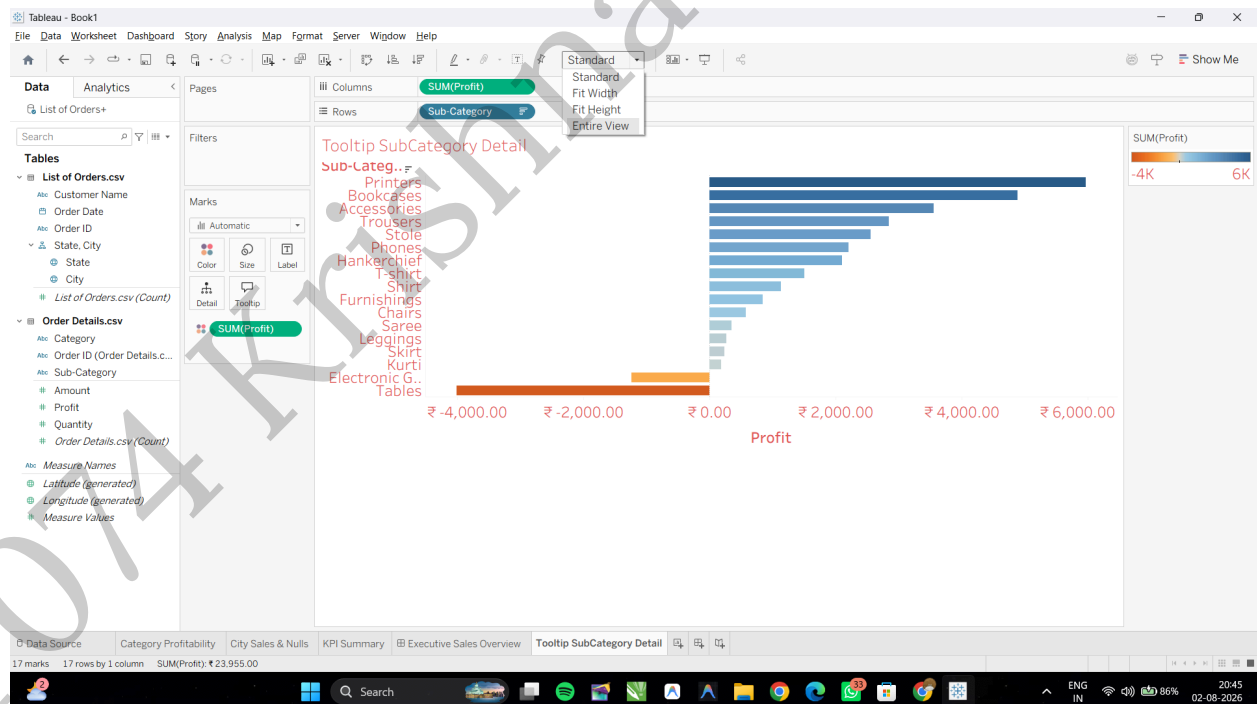
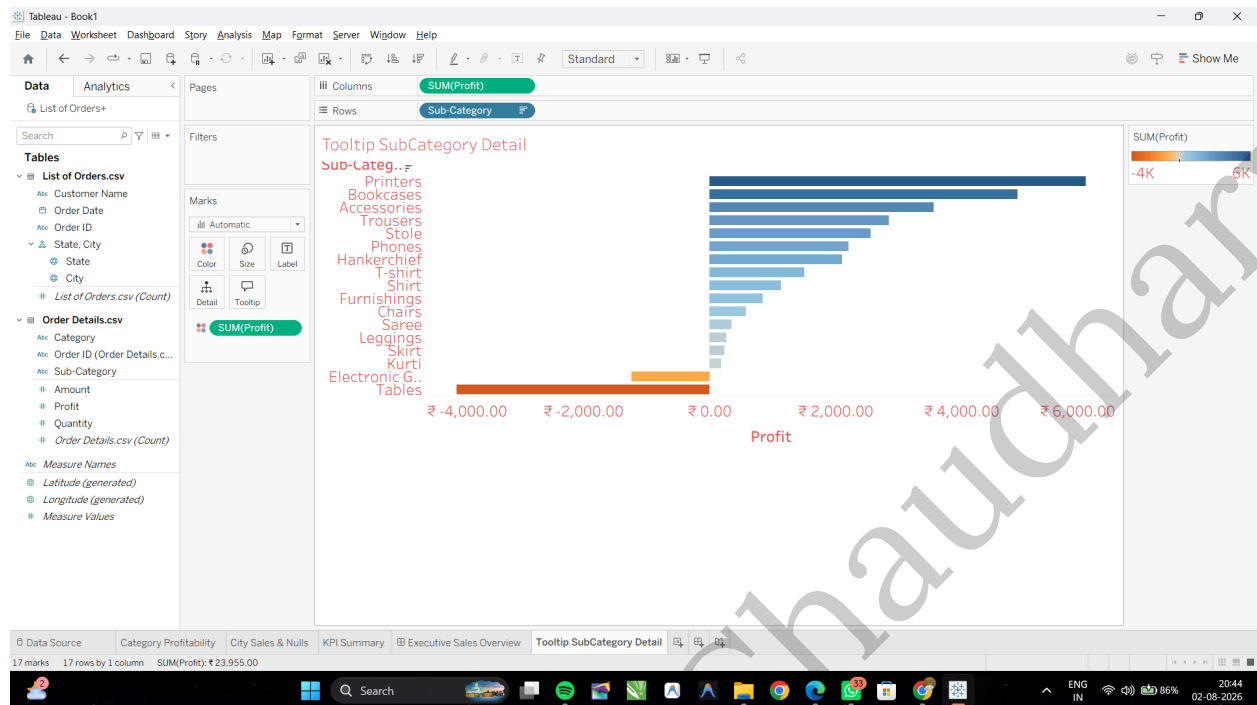
Task 1: Build Secondary Chart

Created a worksheet named **Tooltip SubCategory Detail**, dragged **Sub-Category** to **Rows**, **Profit** to **Columns** and **Color**, sorted the bars in descending order, and changed **Fit** to **Entire View**.



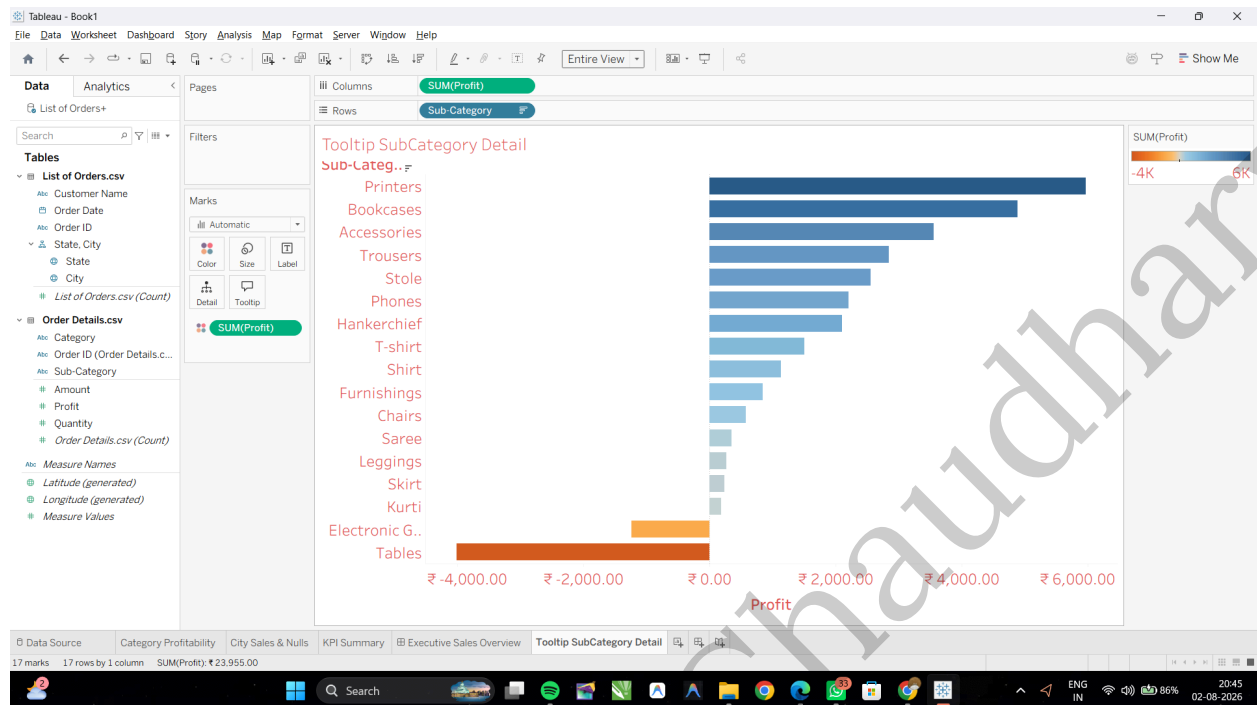
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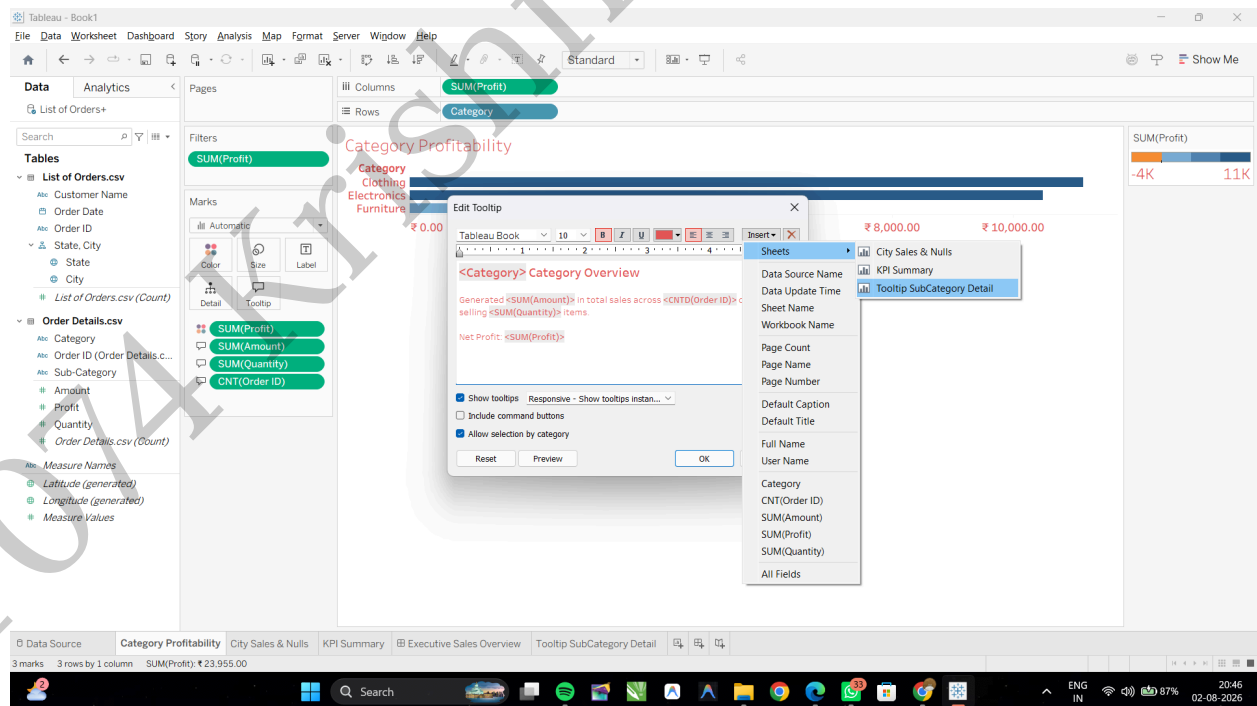
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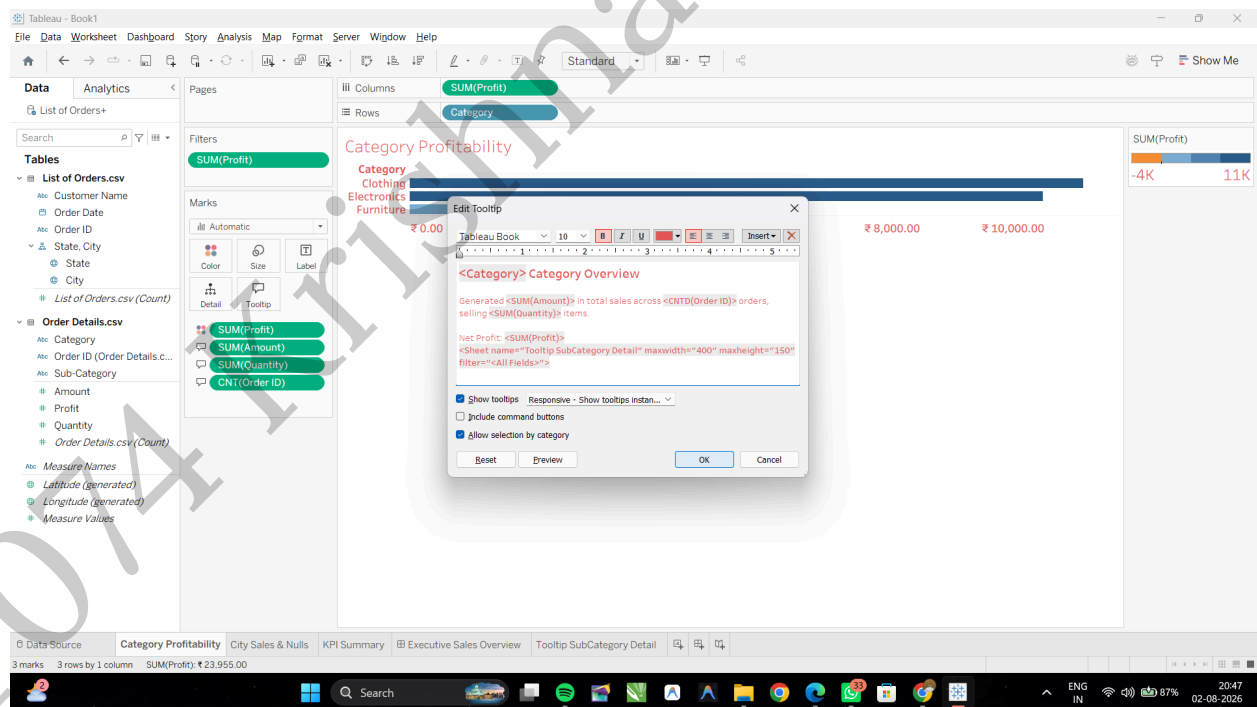
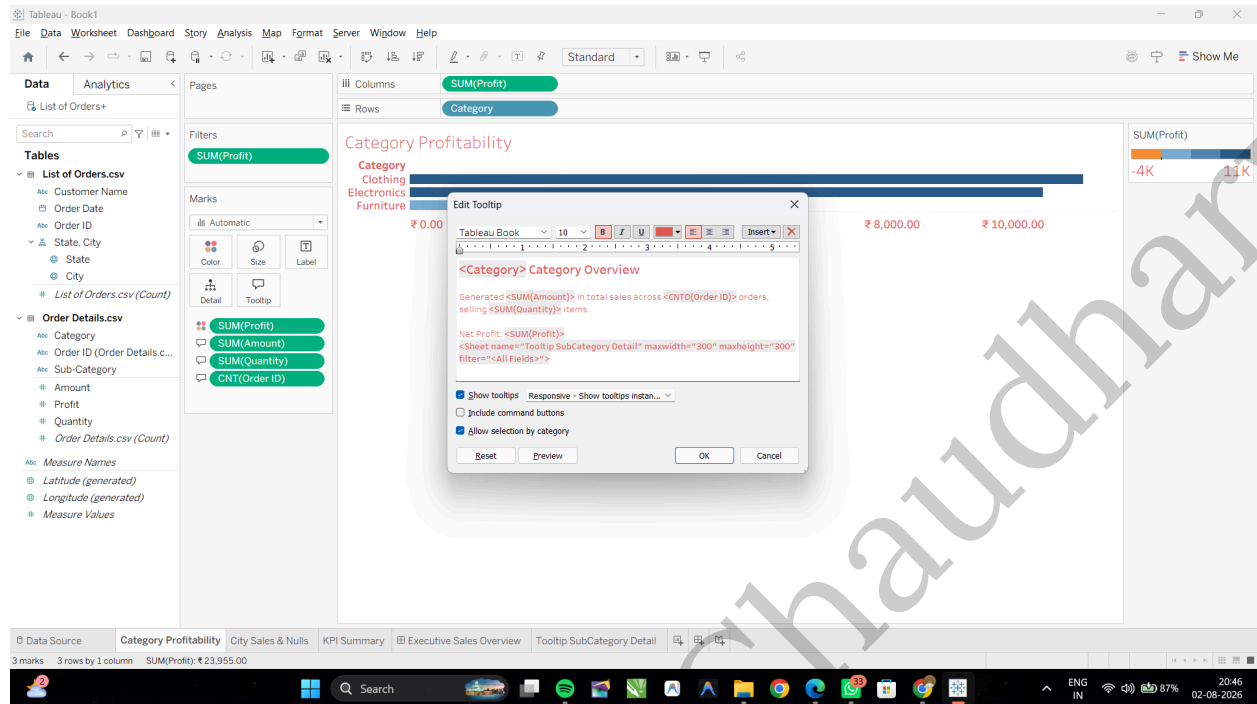
Task 2: Insert Viz into Tooltip

Opened **Category Profitability**, opened the **Tooltip**, inserted **Tooltip SubCategory Detail** using **Insert** → **Sheets**, and adjusted **maxwidth** = 400 and **maxheight** = 200.



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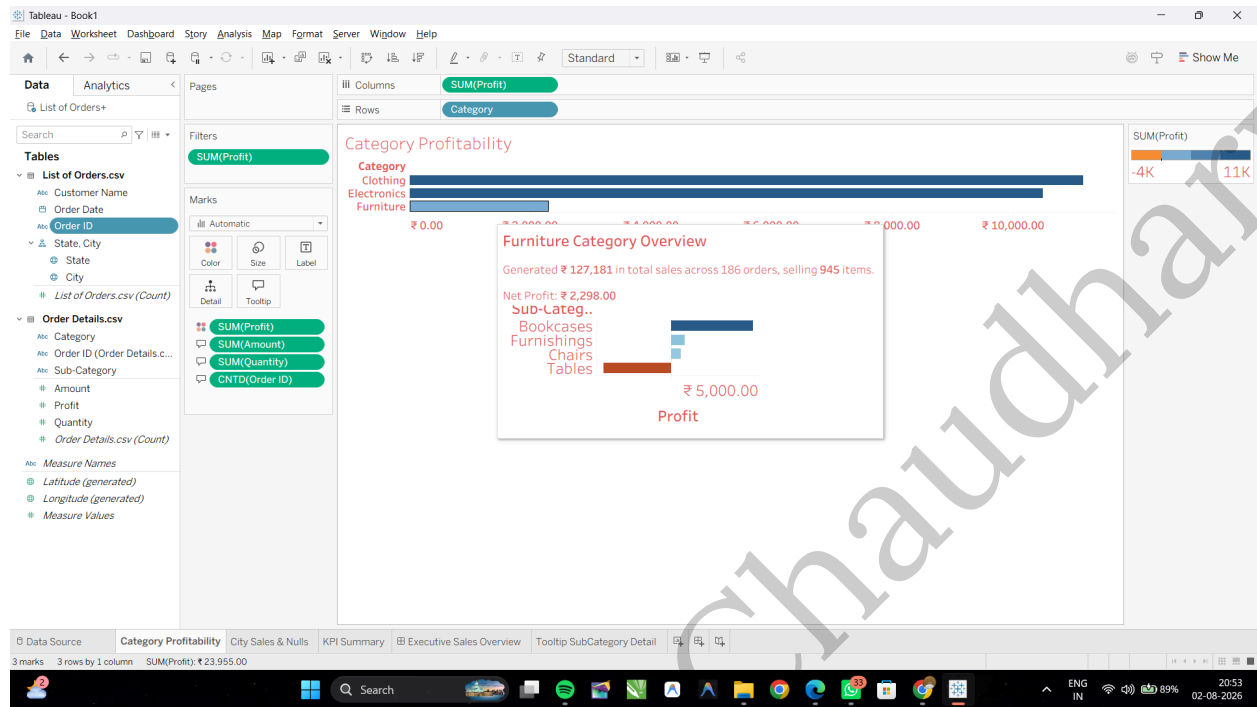


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Task 3: Test Viz in Tooltip

Hovered over the bars and verified the custom tooltip.



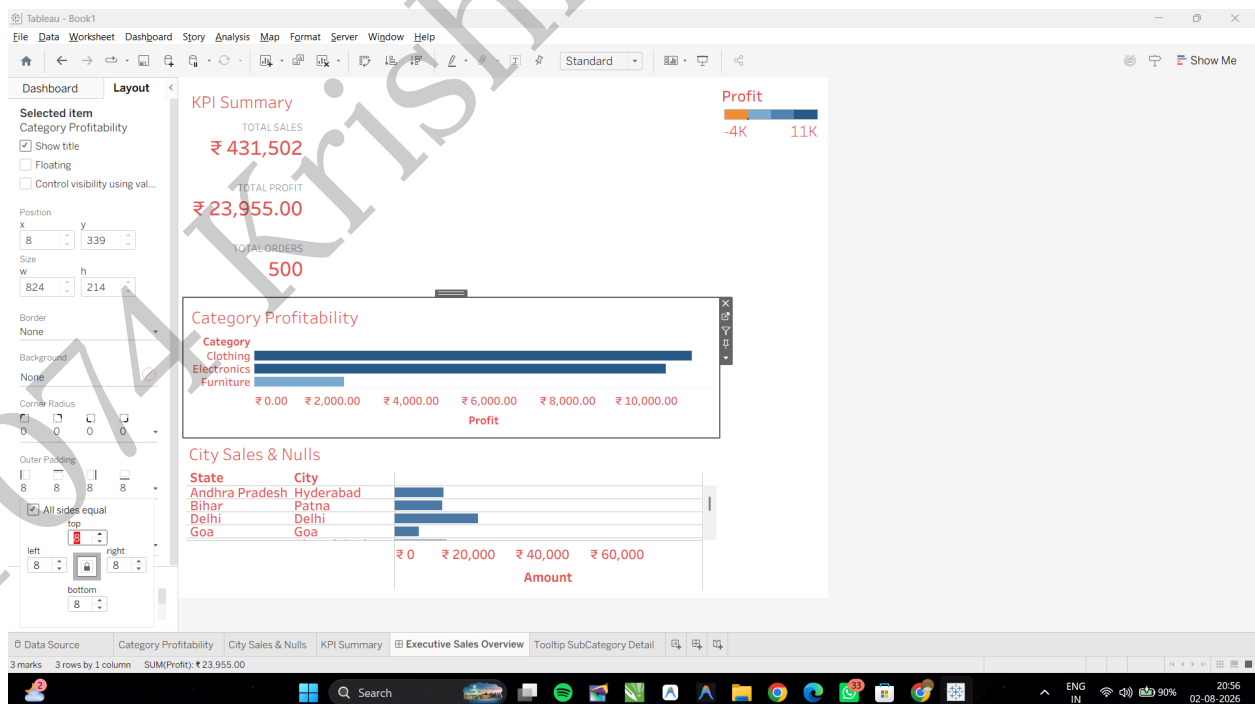
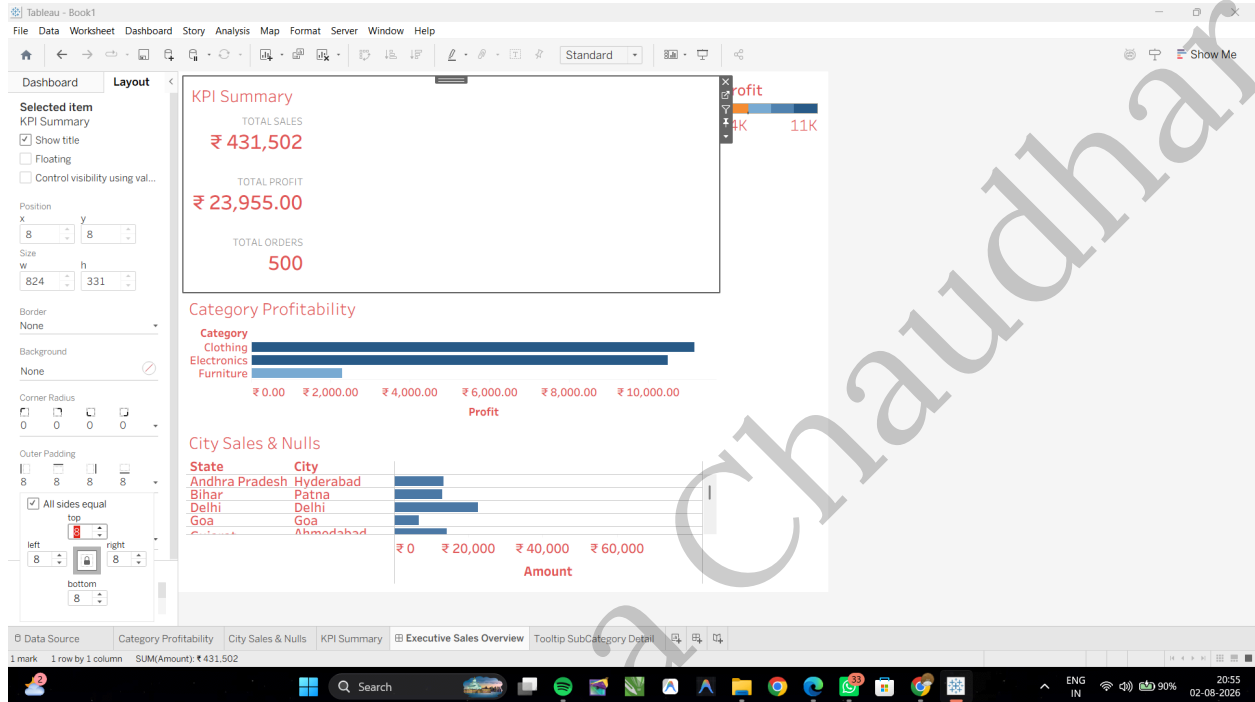
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F: Improve Readability and User Experience

Task 1: Refine Dashboard Padding

Opened the **Executive Sales Overview** dashboard, selected **KPI Summary**, opened the **Layout** pane, applied **8 px Outer Padding**, and repeated the same padding for **Category Profitability** and **City Sales & Nulls**.

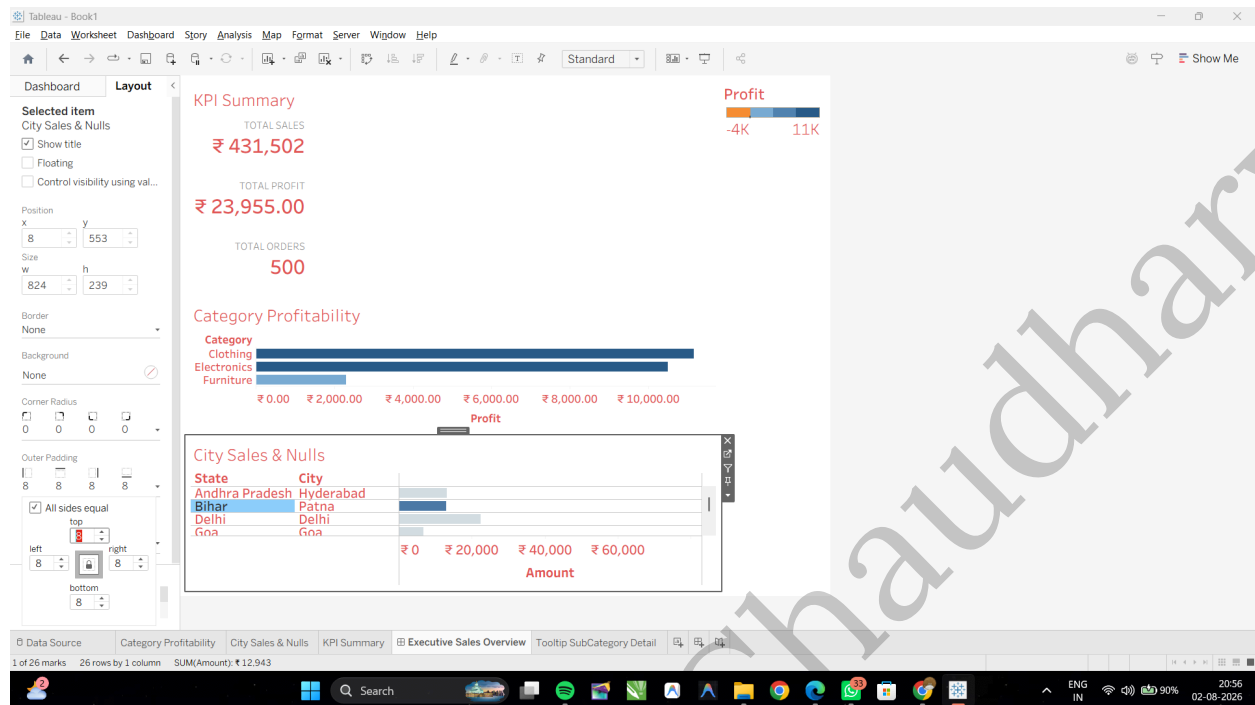


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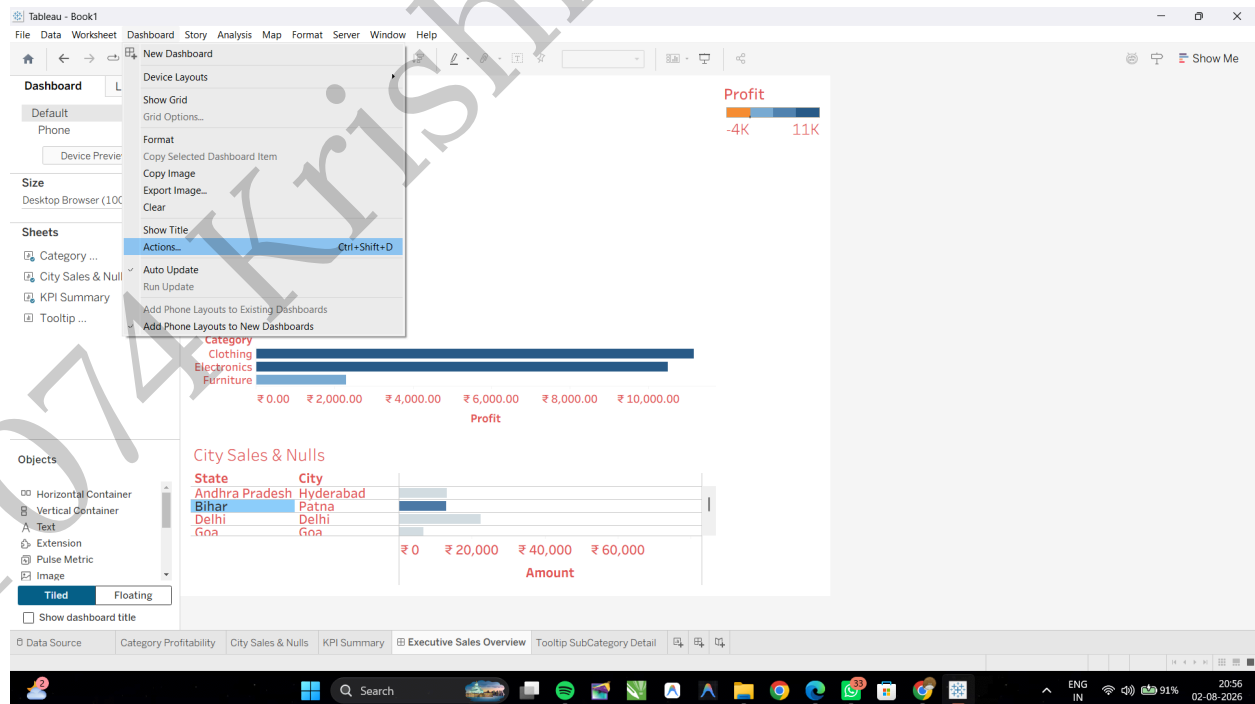
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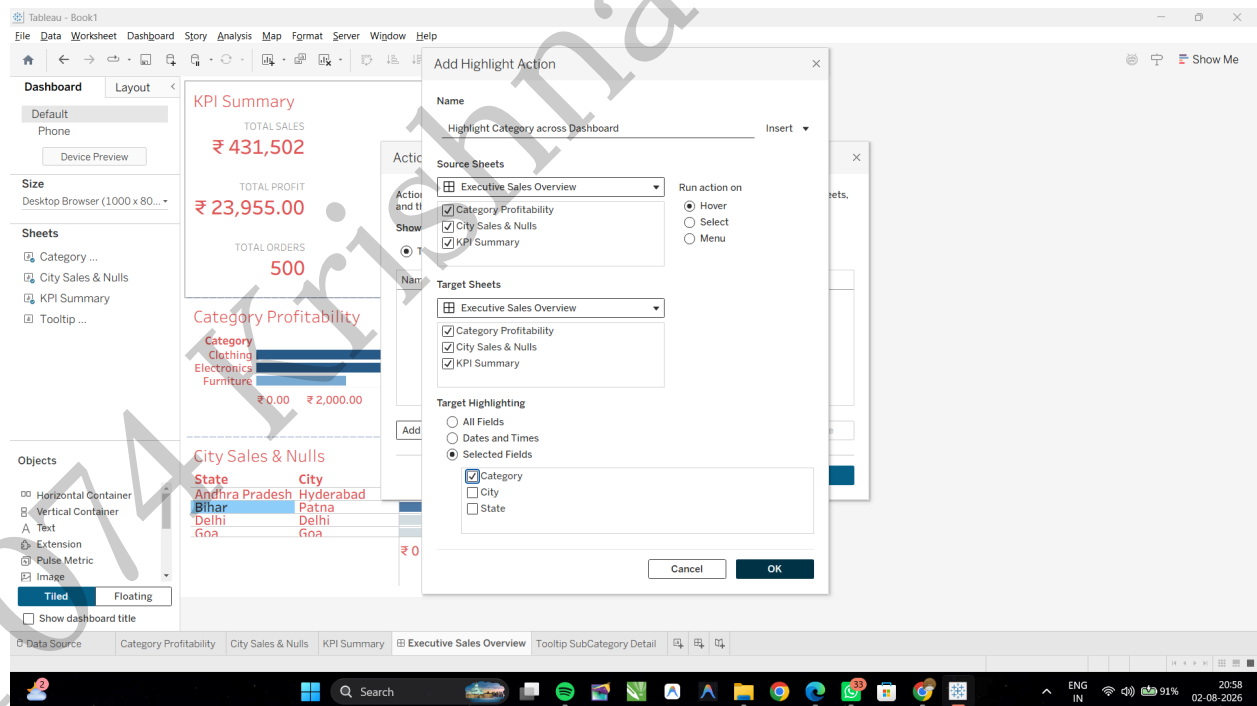
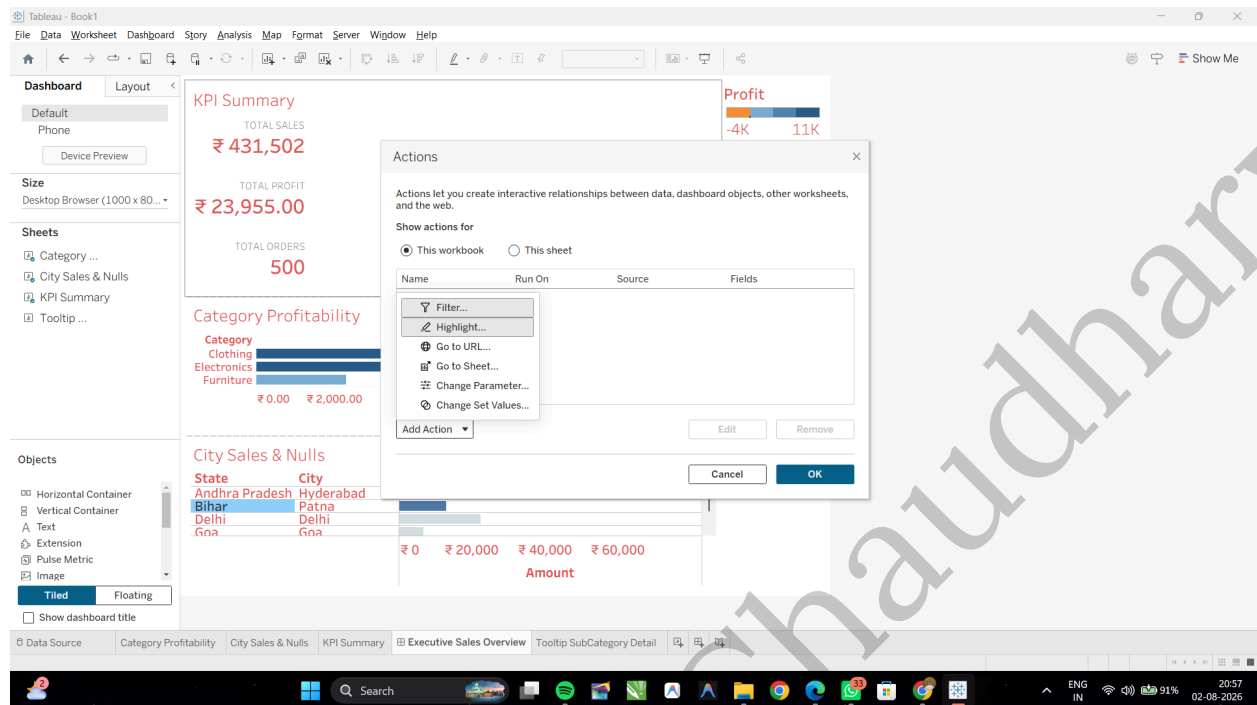


Task 2: Add Highlight Action

Opened Dashboard → Actions, added a Highlight Action, and configured it with the name Highlight Category across Dashboard, Source Sheets as Executive Sales Overview, Target Sheets as All Worksheets, Run Action as Hover, and Target Field as Category.



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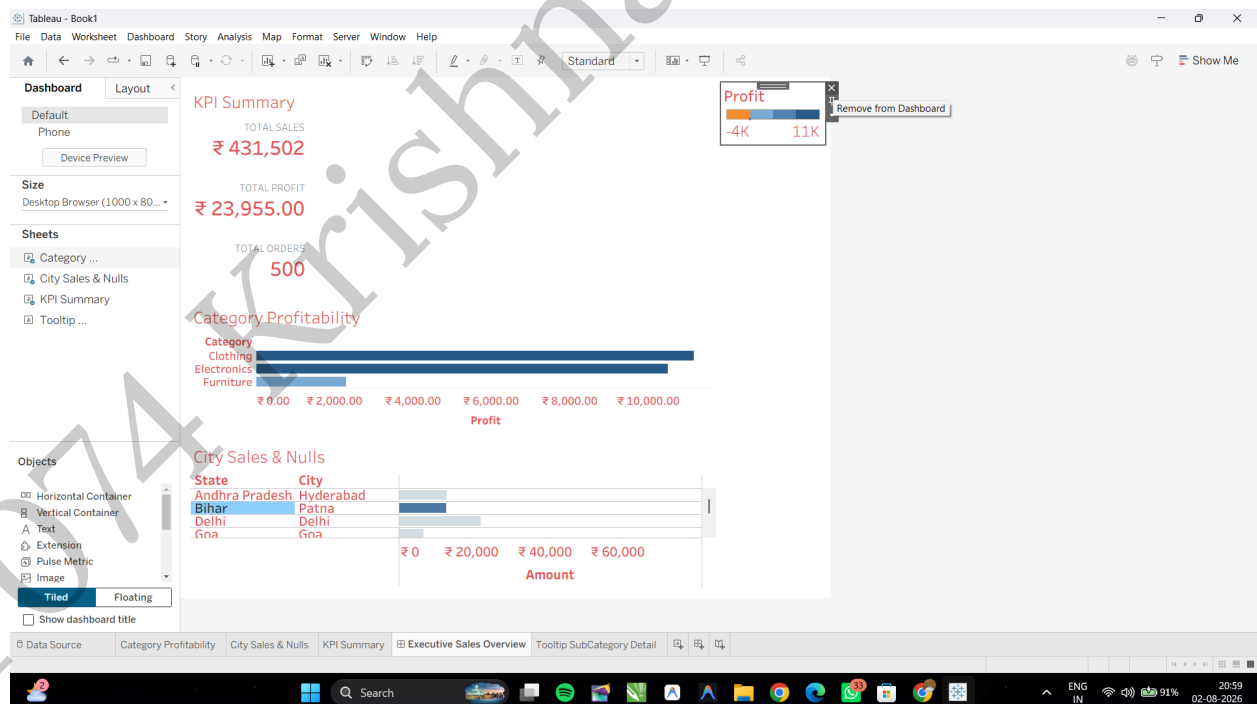
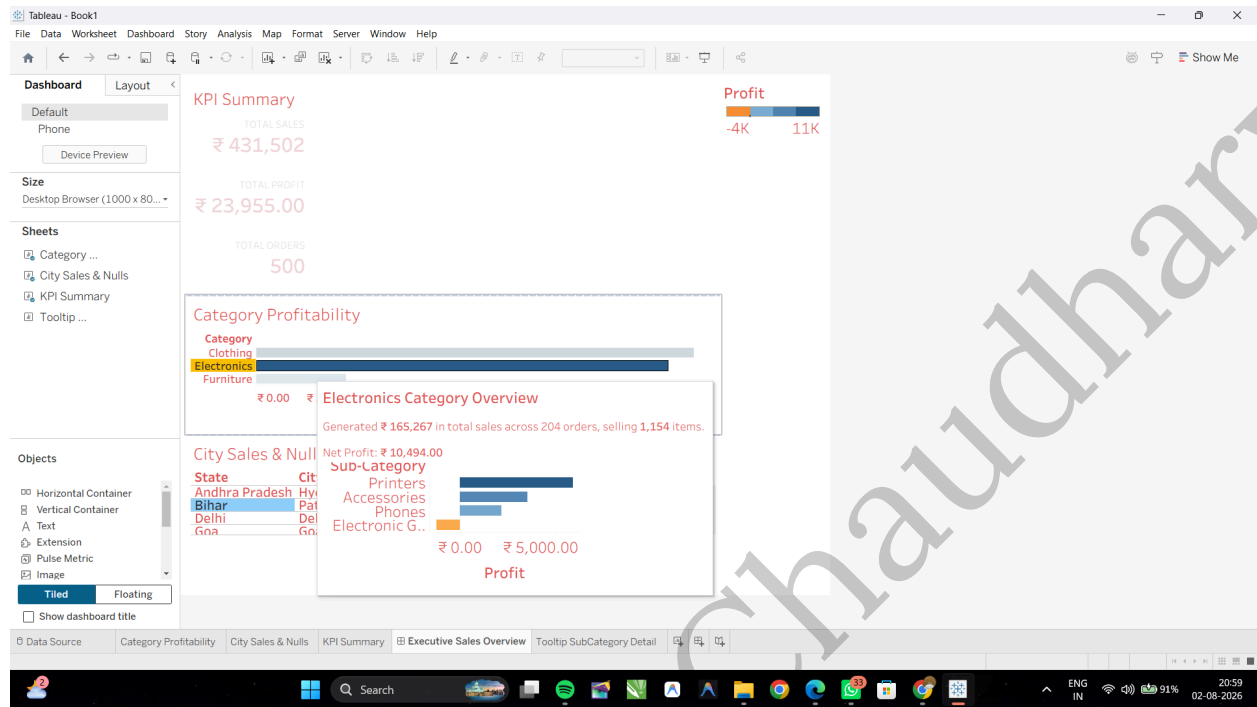


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Task 3: Clean Up Legends

Removed unnecessary legends and positioned the important legends appropriately.



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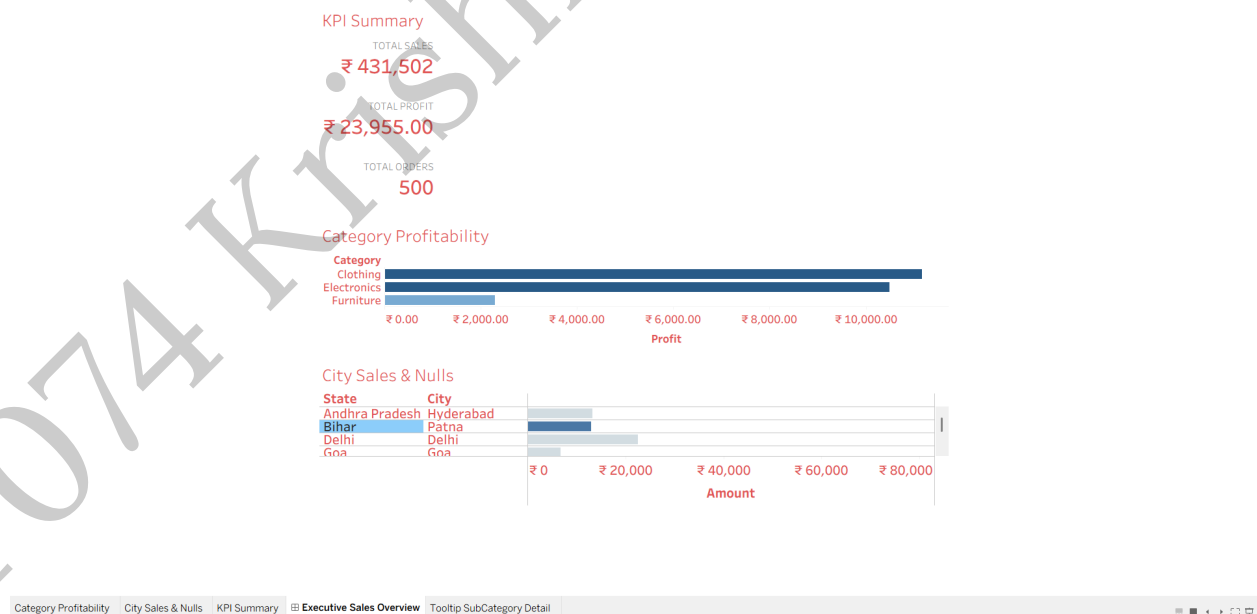
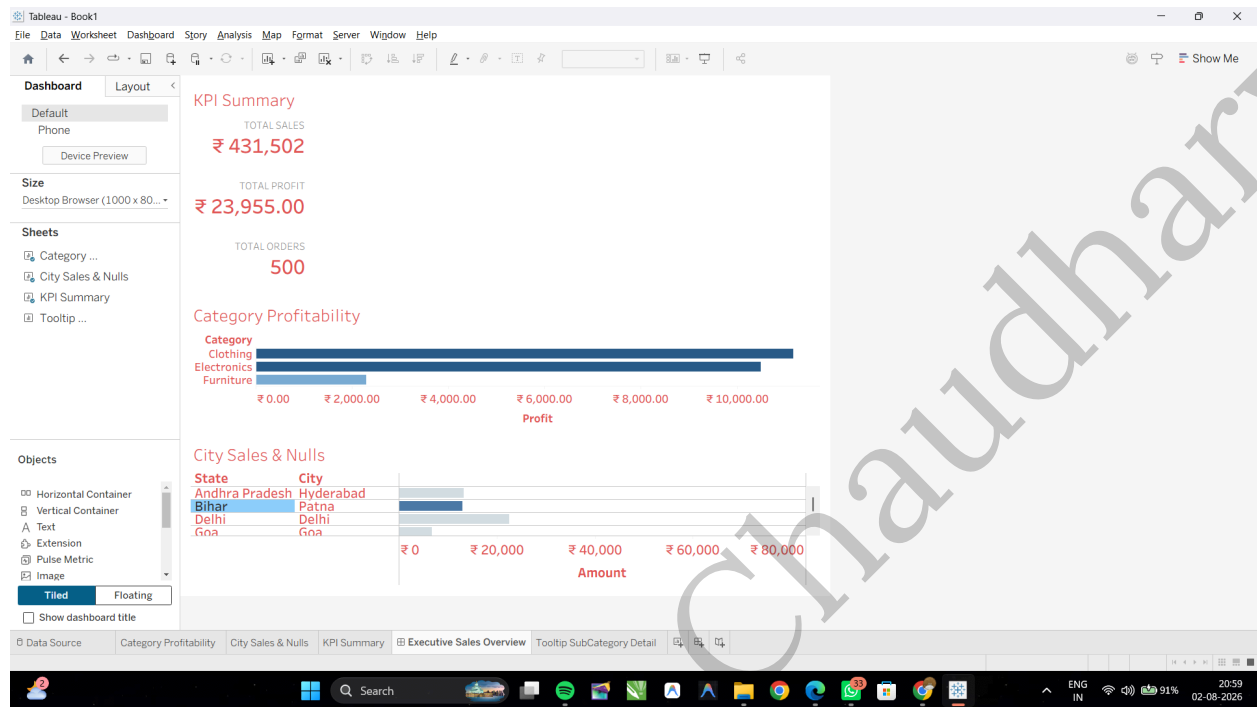
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Task 4: Final Dashboard Check

Verified font consistency, currency formatting, tooltip functionality, Viz in Tooltip, and the Highlight Action.



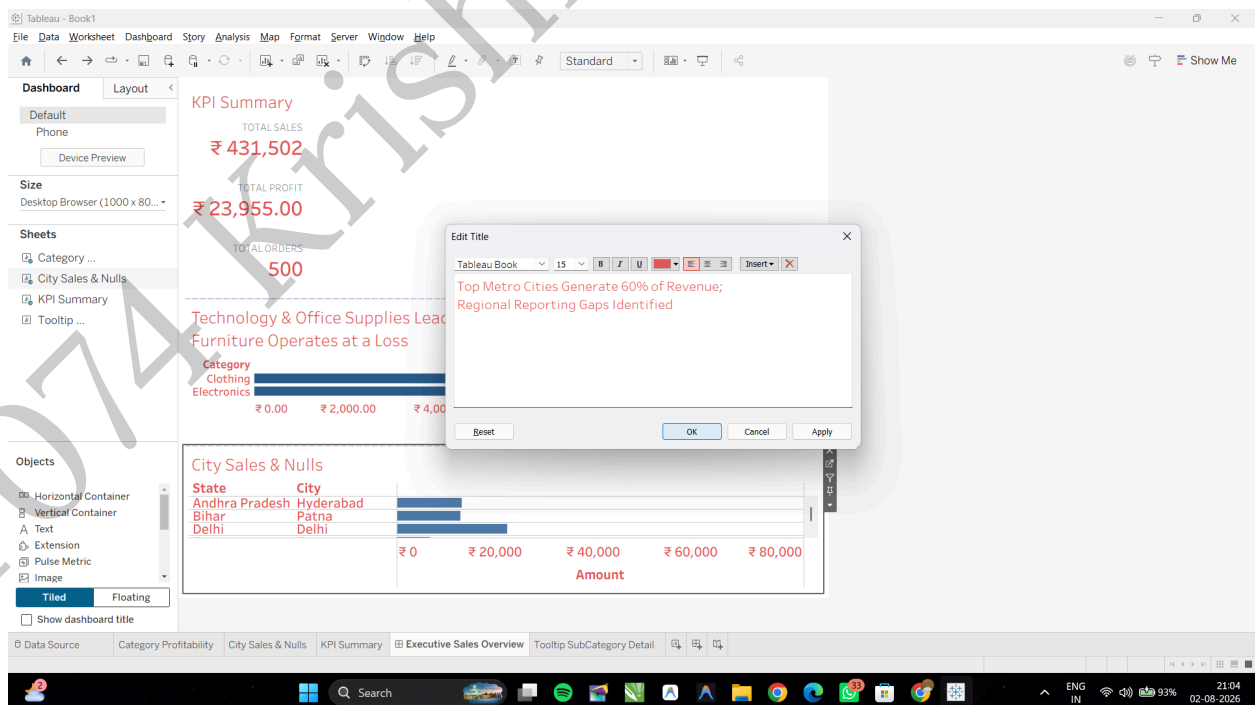
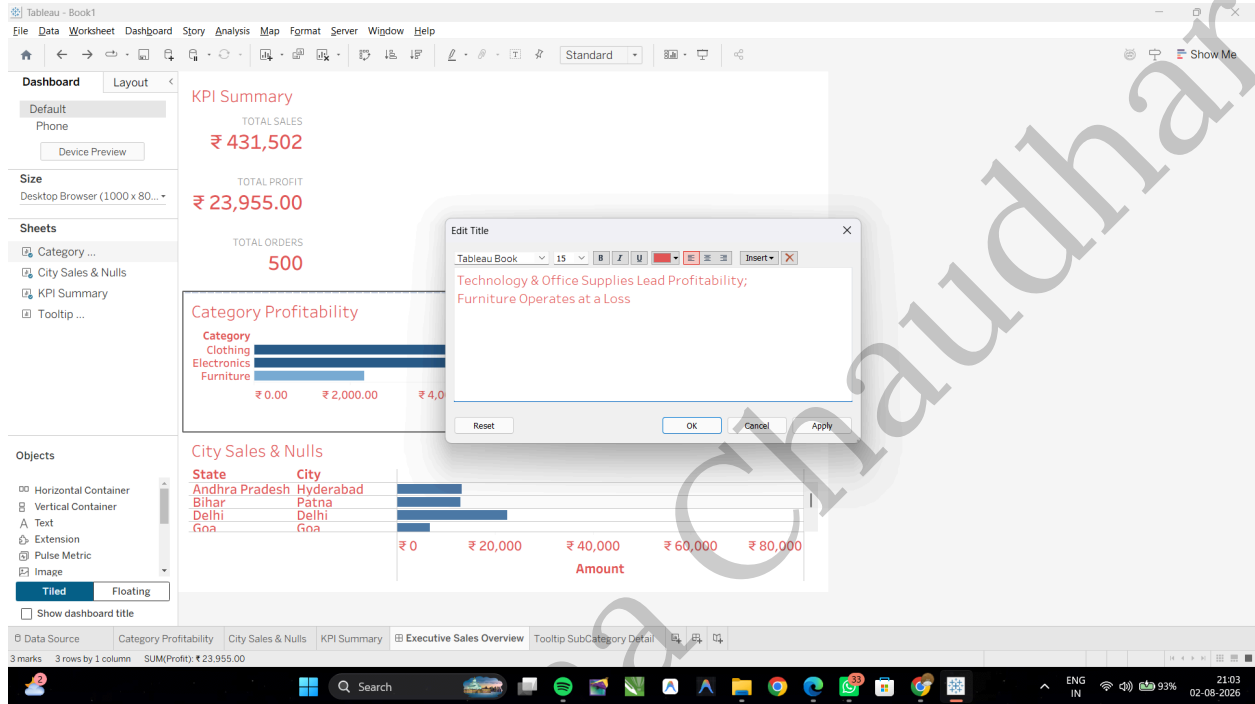
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G: Apply Best Practices for Visual Storytelling

Task 1: Re-structure Dashboard Layout

Arranged the worksheets in an **F-pattern** layout with **KPI Summary**, **Category Profitability**, and **City Sales & Nulls**. And Updated the Category chart title and the City Sales chart title.



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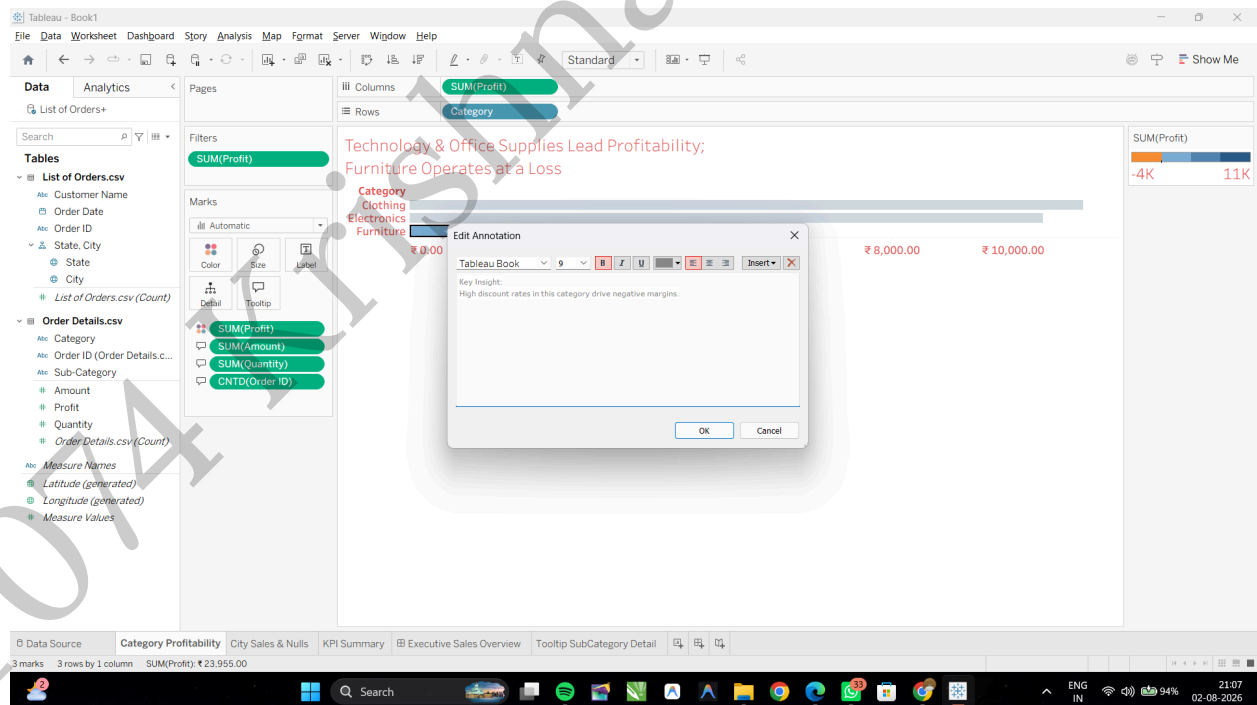
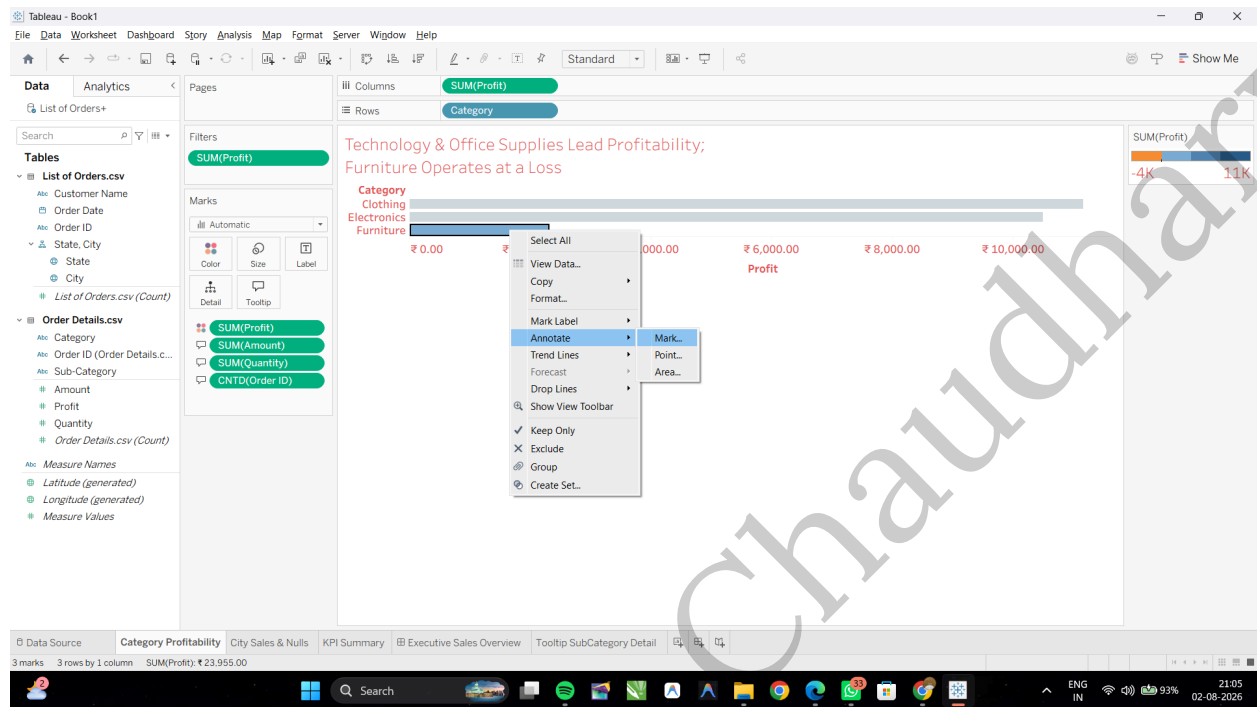
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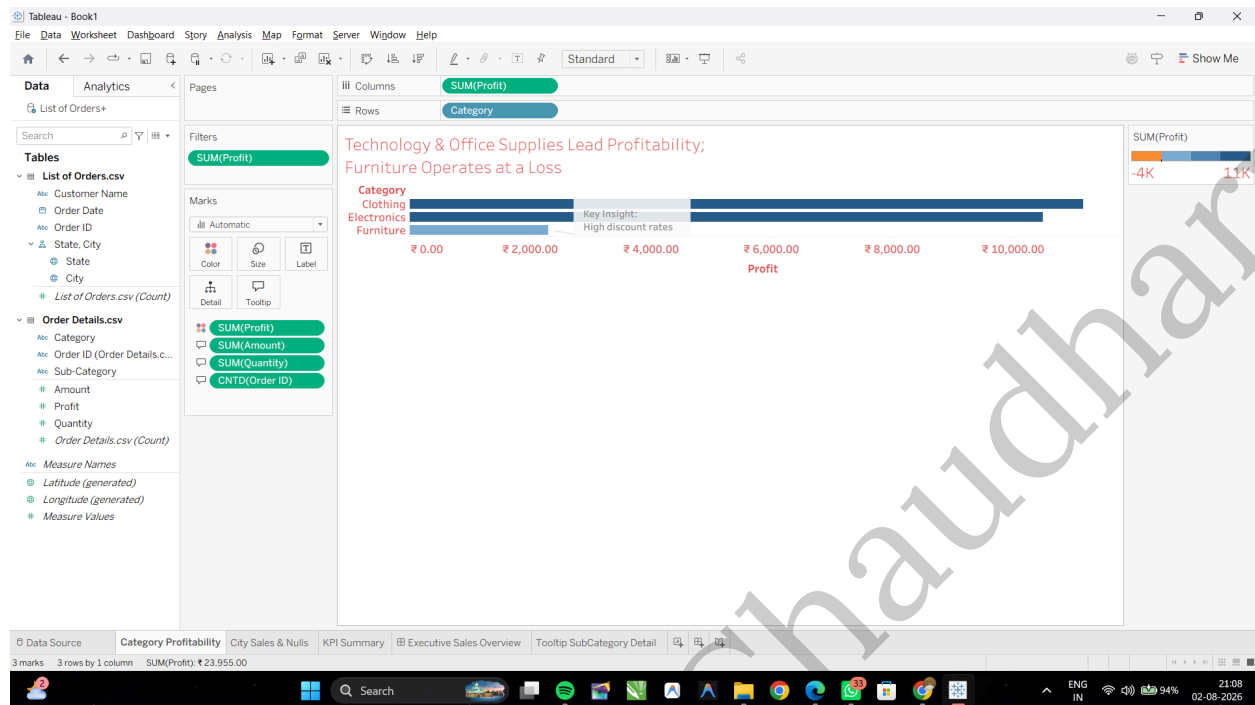
Task 3: Add Annotation

Selected the **Furniture** category, added a **Mark Annotation**, inserted the business insight, formatted the annotation, and positioned it beside the chart.



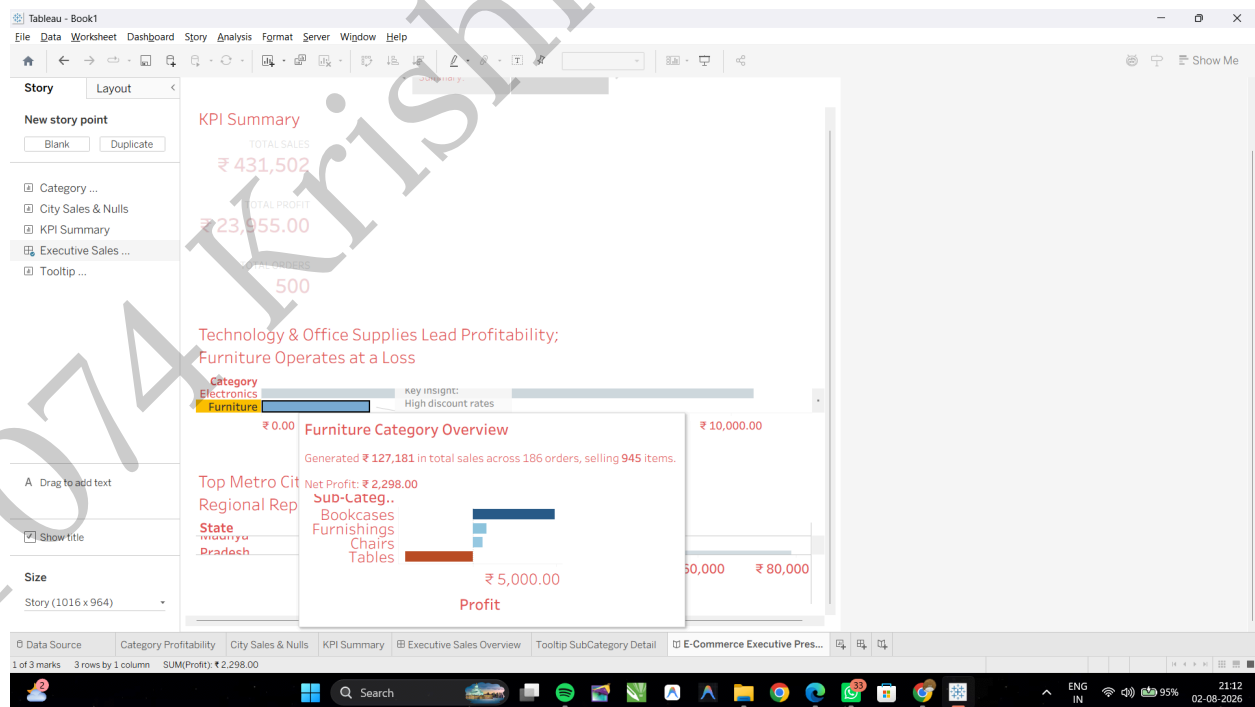
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Task 4: Create Story Points

Created a new Story, renamed it E-Commerce Executive Presentation, added the Executive Sales Overview dashboard, created Story Point 1 titled Executive Summary, added a second story point, inserted the dashboard again, highlighted the Furniture category, created Story Point 2 titled Profit Bottleneck, and saved the workbook.



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SUBJECT NAME: DATA VISUALIZATION WITH POWER BI & TABLEAU

